

Mathematics Department

Mathematics has been described as the queen of all sciences. Understanding mathematics enables one to explain and analyze not only science and nature but almost all disciplines from archeology to zoology. Most recently, mathematics has become an indispensable tool in finance and other business related areas. To ensure that mathematics is available for students with varied backgrounds and different professional goals, the department offers courses at all levels. Advanced courses are designed to be taken by mathematics and actuarial science majors and those in related fields.

Programs

ASCI-BA - Actuarial Science

Overview

Institution Code

BAR01

Official Name of Program

Actuarial Science

Plan Code

ASCI-BA

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

Degree Designation

BA - Bachelor of Arts

CSC-BS - Computer Science

Overview

Institution Code

BAR01

Official Name of Program

Computer Science

Plan Code

CSC-BS

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

Degree Designation

BS - Bachelor of Science

Requirements

Simple Requisites

Degree Requirements - Bachelor of Arts

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 120 credits

AND

Earn at least 30 credits

In Residence at Baruch College

AND

Earn at least 90 credits

Liberal Arts and Sciences Content Courses

AND

Earn a minimum GPA of 2.0

Degree Requirements - Bachelor of Science

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 120 credits

AND

Earn at least 30 credits

In Residence at Baruch College

AND

Earn at least 90 credits

Liberal Arts and Sciences Content Courses

AND

Earn a minimum GPA of 2.0

General Education Requirements - Required Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Required Core English Composition

Complete ALL of the following Courses:

- ENG 2100 - Writing I
OR ENG 2100T - Writing I
OR ENG 2100H - Honors Writing I
- ENG 2150 - Writing II
OR ENG 2150T - Writing II
OR ENG 2150H - Honors - Writing II

General Education Requirements - Required Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Required Core English Composition

Complete ALL of the following Courses:

- ENG 2100 - Writing I
OR ENG 2100T - Writing I
OR ENG 2100H - Honors Writing I
- ENG 2150 - Writing II
OR ENG 2150T - Writing II
OR ENG 2150H - Honors - Writing II

A course in this area must meet all of the following learning outcomes. A student will:

- Read and listen critically and analytically, including identifying an argument's major assumptions and assertions and evaluating its supporting evidence.
- Write clearly and coherently in varied, academic formats (such as formal essays, research papers, and reports) using standard English and appropriate technology to critique and improve one's own and others' texts.
- Demonstrate research skills using appropriate technology, including gathering, evaluating, and synthesizing primary and secondary sources.
- Support a thesis with well-reasoned arguments, and communicate persuasively across a variety of contexts, purposes, audiences, and media.
- Formulate original ideas and relate them to the ideas of others by employing the conventions of ethical attribution and citation.

AND

Required Core Mathematical and Quantitative Reasoning

Complete at least 1 of the following courses:

- MTH 2140 - Mathematics and Quantitative Reasoning
- MTH 2160 - Ideas in Mathematics and Their Applications
- MTH 2003 - Precalculus and Elements of Calculus 1A
- MTH 2009 - Precalculus
- OR MTH 2009T - Precalculus
- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 2610 - Calculus I
- MTH 2001 - Pre-Calculus
- PSY 2100 - Statistics for Social Science
- BIO 2100 - Biostatistics
- OR ENV 2100 - Biostatistics
- STA 2000 - Business Statistics I
- MTH 2001 - Pre-Calculus
- MTH 2600 - Calculus I with Trigonometry Refresher

A course in this area must meet all of the following learning outcomes. A student will:

- Interpret and draw appropriate inferences from quantitative representations, such as formulas, graphs, or tables.
- Use algebraic, numerical, graphical, or statistical methods to draw accurate conclusions and solve mathematical problems.
- Represent quantitative problems expressed in natural language in a suitable mathematical format.
- Effectively communicate quantitative analysis or solutions to mathematical problems in written or oral form.
- Evaluate solutions to problems for reasonableness using a variety of means, including informed estimation.
- Apply mathematical methods to problems in other fields of study.

The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the Mathematical and Quantitative Reasoning requirement of the Pathways curriculum without

A course in this area must meet all of the following learning outcomes. A student will:

- Read and listen critically and analytically, including identifying an argument's major assumptions and assertions and evaluating its supporting evidence.
- Write clearly and coherently in varied, academic formats (such as formal essays, research papers, and reports) using standard English and appropriate technology to critique and improve one's own and others' texts.
- Demonstrate research skills using appropriate technology, including gathering, evaluating, and synthesizing primary and secondary sources.
- Support a thesis with well-reasoned arguments, and communicate persuasively across a variety of contexts, purposes, audiences, and media.
- Formulate original ideas and relate them to the ideas of others by employing the conventions of ethical attribution and citation.

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- OR MTH 2009T - Precalculus
- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 2610 - Calculus I
- MTH 2001 - Pre-Calculus
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- Evaluate solutions to problems for reasonableness using a variety of means, including informed estimation.
- Apply mathematical methods to problems in other fields of study.

The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the Mathematical and Quantitative Reasoning requirement of the Pathways curriculum without

appeal: MTH 2000; MTH 2300; MTH 2300; MTH 2301; MTH 2100; MTH 2150; MTH 2006; MTH 2206; and MTH 2630. MTH 3010 Calculus II is considered a STEM Variant course, and may also be used to satisfy this requirement without appeal.

**Zicklin students should not take MTH 2140 or MTH 2160, which do not meet the mathematics requirement for the BBA degree. Zicklin students must complete a course in precalculus (and must satisfy the prerequisites for whichever precalculus course they take). Therefore, these students usually take MTH 2003 or MTH 2009. Students who enter Baruch with a calculus placement (MTH 2207 or MTH 2610) may take STA 2000 to satisfy their Pathways requirement.

MTH 2140 and MTH 2160 are not appropriate for students within the Weissman School of Arts and Sciences whose major requires a statistics course or additional math courses (such majors include: Actuarial Science, Biological Sciences, Computer Science, Economics, Financial Mathematics, Natural Sciences ad hoc, and Statistics).

AND

Complete at least 1 courses in the following course sets:

- Required Core Life and Physical Sciences

A course in this area must meet all of the following learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a life or physical science.
- Apply the scientific method to explore natural phenomena, including hypothesis development, observation, experimentation, measurement, data analysis, and data presentation.
- Use the tools of a scientific discipline to carry out collaborative laboratory investigations.
- Gather, analyze, and interpret data and present it in an effective written laboratory or fieldwork report.
- Identify and apply research ethics and unbiased assessment in gathering and reporting scientific data.

NOTE: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the **Life and Physical Sciences** or the **Scientific World** requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the **Life and Physical Sciences** or the **Scientific World** requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO 2010 Principles of Biology I
CHM 2003 General Chemistry I
PHY 2003 General Physics I
PHY 3010 Quantitative Physics I

Additional Comments:

The Department of Natural Sciences offers "stand-alone" sections of their Pathways courses (BIO 1011L, BIO 1012, etc.), specifically designed for

appeal: MTH 2000; MTH 2300; MTH 2300; MTH 2301; MTH 2100; MTH 2150; MTH 2006; MTH 2206; and MTH 2630. MTH 3010 Calculus II is considered a STEM Variant course, and may also be used to satisfy this requirement without appeal.

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Additional Comments:

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students who are only required to complete one half of the paired courses – lecture or lab, not both. For example, a student who transfers a **Scientific World** course, may satisfy the **Life and Physical Sciences** requirement with a stand-alone section of a Natural Sciences course in that category. These stand-alone sections are not open to students who enter Baruch as freshmen or to transfer students who have satisfied neither the **Life & Physical Sciences** nor the **Scientific World** requirement at the time they entered Baruch. Please consult the Department of Natural Sciences to request registration permission for any of their stand-alone sections (17 Lexington Ave, room 506; 646-660-6200)

students who are only required to complete one half of the paired courses – lecture or lab, not both. For example, a student who transfers a **Scientific World** course, may satisfy the **Life and Physical Sciences** requirement with a stand-alone section of a Natural Sciences course in that category. These stand-alone sections are not open to students who enter Baruch as freshmen or to transfer students who have satisfied neither the **Life & Physical Sciences** nor the **Scientific World** requirement at the time they entered Baruch. Please consult the Department of Natural Sciences to request registration permission for any of their stand-alone sections (17 Lexington Ave, room 506; 646-660-6200)

General Education Requirements - Flexible Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 courses in the following course sets:

- Flexible Core World Cultures and Global Issues

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring world cultures or global issues, including, but not limited to, anthropology, communications, cultural studies, economics, ethnic studies, foreign languages (building upon previous language acquisition), geography, history, political science, sociology, and world literature.
- Analyze culture, globalization, or global cultural diversity, and describe an event or process from more than one point of view.
- Analyze the historical development of one or more non-U.S. societies.
- Analyze the significance of one or more major movements that have shaped the world's societies.
- Analyze and discuss the role that race, ethnicity, class, gender, language, sexual orientation, belief, or other forms of social differentiation play in world cultures or societies.
- Speak, read, and write a language other than English, and use that language to respond to cultures other than one's own.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core U.S. Experience in Its Diversity

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the U.S. experience in its diversity, including, but not limited to, anthropology, communications, cultural studies, economics, history, political science, psychology, public affairs, sociology, and U.S. literature.
- Analyze and explain one or more major themes of U.S. history from more than one informed perspective.
- Evaluate how indigenous populations, slavery, or immigration have shaped the development of the United States.

General Education Requirements - College Option

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 of the following courses:

- CMP 2800 - Great Works of Literature I
OR ENG 2800 - Great Works of Literature I
- CMP 2850 - Great Works of Literature II
OR ENG 2850 - Great Works of Literature II

AND

Earn at least 3 credits

4000-level CIC "capstone" course for one of the liberal arts minors

This course must be completed at Baruch.

AND

Earn at least 6 credits

Two 3000-level liberal arts course toward the same liberal arts minor

- The liberal arts minor cannot be within the same area of study as the major. This is also true for students who are required to complete part of the minor for the College Option.
- There is no GPA stipulation attached to the College Option courses, but in order to graduate with the liberal arts minor, a student must have a grade point average of at least 2.00 in the three courses that make up his or her liberal arts minor.

Additional Comments:

The College Option consists of as many as four courses (6, 9, or 12 credits), depending on how much coursework the student has completed at another institution.

General Education Requirements - Flexible Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 courses in the following course sets:

- Flexible Core World Cultures and Global Issues

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods

- Explain and evaluate the role of the United States in international relations.
- Identify and differentiate among the legislative, judicial, and executive branches of government and analyze their influence on the development of U.S. democracy.
- Analyze and discuss common institutions or patterns of life in contemporary U.S. society and how they influence, or are influenced by, race, ethnicity, class, gender, sexual orientation, belief, or other forms of social differentiation.

Note: *HIS 1004 History of American Civilization I* is no longer offered at Baruch, but it does appear on students' transfer evaluations. This course may be used to satisfy the U.S. Experience in its Diversity requirement of the Pathways curriculum without appeal.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Creative Expression

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring creative expression, including, but not limited to, arts, communications, creative writing, media arts, music, and theater.
- Analyze how arts from diverse cultures of the past serve as a foundation for those of the present, and describe the significance of works of art in the societies that created them.
- Articulate how meaning is created in the arts or communications and how experience is interpreted and conveyed.
- Demonstrate knowledge of the skills involved in the creative process.
- Use appropriate technologies to conduct research and to communicate.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Individual and Society

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the relationship between the individual and society, including, but not limited to, anthropology, communications, cultural studies, history, journalism, philosophy, political science, psychology, public affairs, religion, and sociology.
- Examine how an individual's place in society affects experiences, values, or choices.
- Articulate and assess ethical views and their underlying premises.
- Articulate ethical uses of data and other information resources to respond to problems and questions.
- Identify and engage with local, national, or global trends or ideologies, and analyze their impact on individual or collective decision-making.

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring world cultures or global issues, including, but not limited to, anthropology, communications, cultural studies, economics, ethnic studies, foreign languages (building upon previous language acquisition), geography, history, political science, sociology, and world literature.
- Analyze culture, globalization, or global cultural diversity, and describe an event or process from more than one point of view.
- Analyze the historical development of one or more non-U.S. societies.
- Analyze the significance of one or more major movements that have shaped the world's societies.
- Analyze and discuss the role that race, ethnicity, class, gender, language, sexual orientation, belief, or other forms of social differentiation play in world cultures or societies.
- Speak, read, and write a language other than English, and use that language to respond to cultures other than one's own.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core U.S. Experience in Its Diversity

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the U.S. experience in its diversity, including, but not limited to, anthropology, communications, cultural studies, economics, history, political science, psychology, public affairs, sociology, and U.S. literature.
- Analyze and explain one or more major themes of U.S. history from more than one informed perspective.
- Evaluate how indigenous populations, slavery, or immigration have shaped the development of the United States.
- Explain and evaluate the role of the United States in international relations.
- Identify and differentiate among the legislative, judicial, and executive branches of government and analyze their influence on the development of U.S. democracy.
- Analyze and discuss common institutions or patterns of life in contemporary U.S. society and how they influence, or are influenced by, race, ethnicity, class, gender, sexual orientation, belief, or other forms of social differentiation.

Note: *HIS 1004 History of American Civilization I* is no longer offered at Baruch, but it does appear on students' transfer evaluations. This course may be used to satisfy the U.S. Experience in its Diversity requirement of the Pathways curriculum without appeal.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Creative Expression

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Scientific World

A course in this area must meet at least three of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the scientific world, including, but not limited to: computer science, history of science, life and physical sciences, linguistics, logic, mathematics, psychology, statistics, and technology-related studies.
- Demonstrate how tools of science, mathematics, technology, or formal analysis can be used to analyze problems and develop solutions.
- Articulate and evaluate the empirical evidence supporting a scientific or formal theory.
- Articulate and evaluate the impact of technologies and scientific discoveries on the contemporary world, such as issues of personal privacy, security, or ethical responsibilities.
- Understand the scientific principles underlying matters of policy or public concern in which science plays a role.

Note: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the *Scientific World* or the *Life and Physical Sciences* requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the *Life and Physical Sciences* or the *Scientific World* requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO2010 Principles of Biology I Principles of Biology I
CHM2003 General Chemistry I General Chemistry I
PHY2003 General Physics I General Physics I
PHY3010 Quantitative Physics I Quantitative Physics I

AND

Complete at least 1 courses in the following course sets:

- Flexible Core - Any Area

Additional Comments:

The Flexible Core is made up of six liberal arts courses, which must be taken in five different categories (numbered 4-8): *4) World Cultures and Global Issues*; *5) U.S. Experience in its Diversity*; *6) Creative Expression*; *7) The Individual and Society*; and *8) Scientific World*.

In fulfilling the six-course requirement, students may not take more than two courses from any one department, discipline, or interdisciplinary field.

Please note that some departments offer courses in more than one

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring creative expression, including, but not limited to, arts, communications, creative writing, media arts, music, and theater.
- Analyze how arts from diverse cultures of the past serve as a foundation for those of the present, and describe the significance of works of art in the societies that created them.
- Articulate how meaning is created in the arts or communications and how experience is interpreted and conveyed.
- Demonstrate knowledge of the skills involved in the creative process.
- Use appropriate technologies to conduct research and to communicate.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Individual and Society

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the relationship between the individual and society, including, but not limited to, anthropology, communications, cultural studies, history, journalism, philosophy, political science, psychology, public affairs, religion, and sociology.
- Examine how an individual's place in society affects experiences, values, or choices.
- Articulate and assess ethical views and their underlying premises.
- Articulate ethical uses of data and other information resources to respond to problems and questions.
- Identify and engage with local, national, or global trends or ideologies, and analyze their impact on individual or collective decision-making.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Scientific World

A course in this area must meet at least three of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the scientific world, including, but not limited to: computer science, history of science, life and physical sciences, linguistics, logic, mathematics, psychology, statistics, and technology-related studies.
- Demonstrate how tools of science, mathematics, technology, or formal analysis can be used to analyze problems and develop solutions.
- Articulate and evaluate the empirical evidence supporting a scientific or formal theory.
- Articulate and evaluate the impact of technologies and scientific discoveries on the contemporary world, such as issues of personal privacy, security, or ethical

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category...

- ANT and SOC courses are offered by the Department of Sociology and Anthropology;
- ART, MSC, and THE courses are offered by the Department of Fine and Performing Arts
- BLS and LTS (formerly HSP) courses are offered by the Department of Black and Latino Studies;
- BIO, ENV, CHM, and PHY courses are offered by the Department of Natural Science; and
- Within the Pathways Flexible Core, POL and PAF (formerly PUB) are considered to be a single field.

All Flexible Core courses must meet the following three learning outcomes. A student will:

- Gather, interpret, and assess information from a variety of sources and points of view.
- Evaluate evidence and arguments critically or analytically.
- Produce well-reasoned written or oral arguments using evidence to support conclusions.

The learning goals or outcomes of each area within the Flexible Core are included above, followed by lists of applicable courses.

Baruch no longer restricts students to one course from any given discipline, department, or interdisciplinary field

General Education Requirements - College Option

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 of the following courses:

- CMP 2800 - Great Works of Literature I
OR ENG 2800 - Great Works of Literature I
- CMP 2850 - Great Works of Literature II
OR ENG 2850 - Great Works of Literature II

AND

Earn at least 3 credits

4000-level CIC "capstone" course for one of the liberal arts minors

This course must be completed at Baruch.

AND

Earn at least 6 credits

Two 3000-level liberal arts courses towards the same liberal arts minor.

Additional Comments:

The College Option consists of as many as four courses (6, 9, or 12 credits), depending on how much coursework the student has completed at another institution.

Degree Requirements - Pre-Weissman Core

Type

Completion Requirement

responsibilities.

- Understand the scientific principles underlying matters of policy or public concern in which science plays a role.

Note: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the *Scientific World* or the *Life and Physical Sciences* requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the *Life and Physical Sciences* or the *Scientific World* requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO2010 Principles of Biology I Principles of Biology I

CHM2003 General Chemistry I General Chemistry I

PHY2003 General Physics I General Physics I

PHY3010 Quantitative Physics I Quantitative Physics I

AND

Complete at least 1 courses in the following course sets:

- Flexible Core - Any Area

Additional Comments:

The Flexible Core is made up of six courses, which must be taken in five different categories (numbered 4-8): **4) World Cultures and Global Issues;** **5) U.S. Experience in its Diversity;** **6) Creative Expression;** **7) The Individual and Society;** and **8) Scientific World.**

In fulfilling the six-course requirement, students may not take more than two courses from any one department, discipline, or interdisciplinary field.

Please note that some departments offer courses in more than one category...

- ANT and SOC courses are offered by the Department of Sociology and Anthropology;
- ART, MSC, and THE courses are offered by the Department of Fine and Performing Arts;
- BLS and LTS (formerly HSP) courses are offered by the Department of Black and Latino Studies;
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All Flexible Core courses must meet the following three learning outcomes. A student will:

- Gather, interpret, and assess information from a variety of sources and points of view.
- Evaluate evidence and arguments critically or analytically.
- Produce well-reasoned written or oral arguments using evidence to support conclusions.

The learning goals or outcomes of each area within the Flexible Core are included above, followed by lists of applicable courses.

Baruch no longer restricts students to one course from any given discipline.

Fulfill ALL of the following requirements:

Complete ALL of the following Courses:

- COM 1010 - Speech Communication
OR COM 1010H - Honors Speech Communications

AND

Foreign Language

Earn at least 6 credits

Students may not use proficiency and/or exemption exams to satisfy this requirement.

1st Foreign Language course (based on placement) and 2nd Foreign Language course (the same language as the 1st Foreign Language course)

Students who transfer language credits for languages not taught at Baruch College may use these credits for the foreign language requirement at Baruch College.

Placement Information:

Students should review the guidelines below, and see a faculty advisor in the Department of Modern Languages and Comparative Literature to confirm proper placement if they intend to continue a language to which they have prior exposure. Students have the option of taking a placement exam to ensure that they are placed at the correct level.

Please consult the [Department of Modern Languages and Comparative Literature](#) for assistance

Major Requirements - Overall

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 49 credits

AND

Earn at least 16 credits in residence

AND

Earn a minimum GPA of 2.0

Major Requirements - Program Prerequisites

Type

Prerequisite

Fulfill ALL of the following requirements:

Complete at least 1 of the following:

Complete ALL of the following Courses:

- MTH 3010 - Calculus II
- MTH 3020 - Calculus III

Students may also receive credits to the degree program from a department, or interdisciplinary field

Major Requirement: CS with a Concentration in Environmental Science

Type

Completion Requirement

Complete ALL of the following Courses:

- ENV 1003 - Fundamentals of Ecology
- ENV 1004 - Fundamentals of Ecological Research
- ENV 3016 - Environmental Modeling
- ENV 4900 - Environmental Sciences Capstone

- The environmental science faculty mentor is Dr. Stephen Gosnell.
- This concentration requires two CS electives. Students with this concentration are encouraged to take the Machine learning elective (MTH 4330).

Major Requirement: CS with Concentration in Applied Linguistics

Type

Completion Requirement

Complete ALL of the following Courses:

- ENG 3700 - Introduction to Linguistics: The Study of Language
OR ENG 3750 - The Structure and History of English
- ENG 3960 - Topics in Language

- This concentration focuses on Computer Assisted Language Learning (CALL). The linguistics faculty mentor is Dr. Brooke Schreiber.
- This concentration requires two CS electives

Major Requirements - Overall

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 43 credits

Pathways requirements, consisting of 13 credits of Required Core (including a STEM alternative MTH 2610), 18 credits of Flexible Core, and 12 credits of the College Option

AND

Earn at least 9 credits

Pre-Weissman core courses (including foreign languages)

AND

Earn at least 26 credits

Required Mathematics courses

AND

Earn at least 15 credits

Major/Concentration elective courses

OR

Complete ALL of the following Courses:

- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus
- MTH 3050 is not open to students who completed MTH 3020, MTH 3030, MTH 3035, or their equivalents.

OR

Complete ALL of the following Courses:

- MTH 2610 - Calculus I
- MTH 3010 - Calculus II
- MTH 3020 - Calculus III

OR

Complete ALL of the following Courses:

- MTH 2610 - Calculus I
- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus

OR

Complete ALL of the following Courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 3006 - Elements of Calculus II
- MTH 3030 - Elements of Calculus III

OR

Complete ALL of the following Courses:

- MTH 2207 - Elements of Calculus I and Matrix Algebra
- OR MTH 2600 - Calculus I with Trigonometry Refresher
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3020 - Calculus III

OR

Complete ALL of the following Courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3020 - Calculus III

OR

Complete ALL of the following Courses:

- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3020 - Calculus III

AND

Earn at least 27 credits

Elective courses

Major Requirements - Program Requirements

Type

Completion Requirement

Students must satisfy the following:

1. MTH1023 Int and College Algebra or MTH1030 College Algebra or FSPM1031 College Algebra or placement higher than MTH1030 College Algebra.
2. MTH2000 Pre-Calculus: Coll Alg & Trig or MTH2001 Pre-Calculus or MTH2003 Precal & Elem of Cal 1A or MTH2009 Precalculus or MTH2009T Precalculus or placement of MTH2207 Elements of Calculus I and Ma or higher.
3. MTH2301 Concepts Of Discrete Math. This prerequisite is waived if a student satisfies:
 - Combined GPA of at least 3.5 in Calculus I (MTH2205 Precal and Elements of Cal 1B, MTH2206 Applied Calculus, MTH2207 Elements of Calculus I and Ma, MTH2600 Calculus I Tri Refresher, or MTH2610 Calculus I) and Calculus II (MTH2630 Analy Geom & Calc I, MTH3006 Elements of Calculus II, or MTH3010 Calculus II); OR
 - BC Calculus AP Exam with a score of 4 or 5; OR
 - AB Calculus AP Exam with a score of 4 or 5 and Calculus II (MTH 3010) with a B+ or higher

Students need the equivalent of two semesters of calculus, which may be satisfied by one of the following four options:

Fulfill ANY of the following requirements:

Option 1

- MTH2610 Calculus I- Calculus I OR Calculus AP Exam (AB) with a score of 4 or 5 (transfers to Baruch as MTH 2610)
- MTH3010 Calculus II- Calculus II

OR

Optional 2

Complete ALL of the following Courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
- OR MTH 2206 - Applied Calculus
- OR MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 3006 - Elements of Calculus II

OR

Option 3

Calculus AP Exam (BC) with a score of 4 or 5 (transfers to Baruch as MTH 3010)

OR

OR

Complete ALL of the following Courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3050 - Calculus III and Vector Calculus

OR

Complete ALL of the following Courses:

- MTH 2630 - Analytic Geometry and Calculus I
- MTH 3030 - Elements of Calculus III

AND

Complete ALL of the following Courses:

- ACC 2101 - Principles of Accounting
OR ACC 2101H - Honors Principle of Accounting I
- ECO 1001 - Micro-Economics
OR ECO 1001H - Honors - Micro Economics
- ECO 1002 - Macro-Economics
OR ECO 1002H - Honors - Macro Economics
- FIN 3000 - Principles of Finance
OR FIN 3000H - Honors - Principles of Finance
- FIN 3610 - Corporate Finance
OR FIN 3610H - Honors Corporate Finance

Major Requirements - Required Courses

Type

Completion Requirement

Complete ALL of the following Courses:

- MTH 3300 - Algorithms, Computers, and Programming I
- MTH 4120 - Introduction to Probability
- MTH 4410 - Theory of Interest
- Students who have completed MTH3120 Elementary Probability cannot enroll in MTH4120 Introduction to Probability. They must satisfy the probability requirement by registering for MTH4119 Multivariate Prob Dist as an independent study.
- ECO3100 Intermediate Micro-Economics
Intermediate Micro-Economics and ECO3200 Intermediate Macro-Economics Intermediate Macro-Economics are recommended, but not required.

Complete at least 3 of the following courses:

- MTH 4420 - Actuarial Mathematics I
- MTH 4421 - Actuarial Mathematics II
- MTH 4451 - Short-Term Insurance Mathematics
- MTH 4452 - Short-Term Insurance Mathematics II
- MTH 4430 - Mathematics of Inferential Statistics

Option 4

Complete ALL of the following Courses:

- MTH 2630 - Analytic Geometry and Calculus I

CS+ majors need the equivalent of two semesters of calculus which could be satisfied by one of the following 4 options.

Additional Comments:

Students must meet the following requirement to gain admission to the program:

- In Options 1,2, and 4, the combined calculus GPA must be at least 3.5 out of 4.0.
- MTH3300 (or CIS 2300) and MTH3150 with a minimum grade of B in each course.

Students who fail to obtain sufficiently high grades to enter the program could use the CS courses they took to obtain a CS minor. This minor already exists at Baruch. It consists of MTH3300(or CIS 2300), MTH3150, and one capstone course. Thus, after taking the prerequisites for the CS+ program, a student would need only a capstone course to compete the CS minor.

Major Requirements - Required Courses

Type

Completion Requirement

Complete ALL of the following Courses:

- MTH 3150 - Discrete Math: An Invitation to Computer Science
- MTH 3300 - Algorithms, Computers, and Programming I
OR CIS 2300 - Programming and Computational Thinking
- MTH 4300 - Algorithms, Computers and Programming II
- MTH 4320 - Data Structures and Algorithms
- MTH 4350 - Computer Architecture
- MTH 4355 - Operating Systems
- MTH 4360 - Complexity and Computational Models

Major Requirements- Pre- Weissman Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete ALL of the following Courses:

- COM 1010 - Speech Communication
OR COM 1010H - Honors Speech Communications

AND

Foreign Language

Earn at least 6 credits

Students may not use proficiency and/or exemption exams to satisfy this requirement.

1st Foreign Language course (based on placement) and 2nd Foreign Language course (the same language as the 1st Foreign Language course)

Students who transfer language credits for languages not taught at Baruch College may use these credits for the foreign language requirement at Baruch College.

Major Requirements - Elective Courses

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 of the following courses:

- MTH 4115 - Numerical Methods for Differential Equations in Finance
- MTH 4125 - Introduction to Stochastic Processes
- MTH 4130 - Mathematics of Data Analysis
- MTH 4135 - Computational Methods in Probability
- MTH 4420 - Actuarial Mathematics I
- MTH 4421 - Actuarial Mathematics II
- MTH 4430 - Mathematics of Inferential Statistics
- MTH 4451 - Short-Term Insurance Mathematics
- MTH 4452 - Short-Term Insurance Mathematics II
- MTH 4500 - Introductory Financial Mathematics
- MTH 4600 - Data Analysis and Simulation for Financial Engineers
- MTH 5500 - Stochastic Calculus for Finance

AND

Earn at least 3 credits from the following:

- Liberal Arts Electives
- You may need to take additional Liberal Arts Courses to complete the required credits for your degree.

Placement Information:

Students should review the guidelines below, and see a faculty advisor in the Department of Modern Languages and Comparative Literature to confirm proper placement if they intend to continue a language to which they have prior exposure. Students have the option of taking a placement exam to ensure that they are placed at the correct level.

Please consult the [Department of Modern Languages and Comparative Literature](#) for assistance.

Major Requirements - Elective Courses

Type

Completion Requirement

Fulfill ANY of the following requirements:

Complete ANY of the following Courses:

- CIS 3400 - Database Management Systems
- CIS 3500 - Computer Networking
- CIS 3630 - Principles of Web Design
- CIS 4160 - Web Engineering
- CIS 4560 - Ethical Hacking
- MTH 4135 - Computational Methods in Probability
- MTH 4140 - Graph Theory
- MTH 4150 - Combinatorics
- MTH 4250 - Introduction to Cryptography
- MTH 4322 - Computer Science Internship Seminar
- MTH 4325 - Programming Languages
- MTH 4330 - Introduction to Machine Learning
- MTH 4333 - Advanced Topics in Machine Learning
- MTH 4335 - Introduction to Natural Language Processing
- MTH 4370 - Software Design
- MTH 4375 - Applied Data Privacy
- MTH 4380 - Computer and Network Security
- MTH 4395 - Special Topics in Computer Science
- AAS 3080 - A Survey of Asian History

The program also includes advanced CS elective courses. CS+ majors are required to take 1-2 CS electives, depending on their concentration. A CS+ major with no concentration is required to take four CS electives.

Major Requirements- General Track

Type

Completion Requirement

Computer Science - General Track

Concentration Requirements: 12-16 credits

Students must complete four courses from the computer science electives list.

Major Requirement: CS with Concentration in Biology

Type

Completion Requirement

Complete ALL of the following Courses:

- BIO 2100 - Biostatistics

OR ENV 2100 - Biostatistics

- This concentration focuses on bioinformatics. The biology faculty mentors are Dr. Zachary Calamari and Dr. Rebecca Spokony.
- Students are also required to take Bioinformatics (4 credits), this is a new course which is currently being created by Zachary Calamari.
- This concentration requires two CS electives.

Complete ALL of the following:

Major Requirement: CS with Concentration in Physics

Type

Completion Requirement

Complete ALL of the following Courses:

- PHY 3004 - Physics on the Computer with Python
- PHY 4004 - Statistical Physics with Applications to Mathematical Finance
- This concentration is focused on computational physics. The physics faculty mentors are Dr. Adrian Dumitru and Dr. Stefan Bathe.
- This concentration requires two CS electives.
- Note: PHY 3004 has no significant overlap with the CS Python course MTH 3300 (or CIS 2300)

Major Requirement: CS with Concentration in Financial Mathematics

Type

Completion Requirement

Concentration Requirements: 19-21 crs

- Prerequisite MTH course:, MTH 3030, or MTH 3050
- Computer Science Elective Courses: One course from the computer science electives list
- Financial Mathematics Elective Courses: Three courses from below

Complete ALL of the following Courses:

- MTH 4120 - Introduction to Probability
- MTH 4500 - Introductory Financial Mathematics
- MTH 4115 - Numerical Methods for Differential Equations in Finance
- This concentration is focused on computational finance. The mathematical finance faculty mentor is Dr. Douglas Howard.
- This concentration requires one CS elective. Students with this concentration are encouraged to take the Machine learning elective (MTH 4330).
- MTH 4115 has linear algebra as a pre or co-requisite. However, MTH 3150 would be an alternative prerequisite.

Major Requirement: CS with a Concentration in Psychology

Type

Completion Requirement

Complete ALL of the following Courses:

- PSY 4004 - Last Gap Psychology

- PSY 1001L - Lect Gen Psychology
- PSY 3001 - Research Methods in Psychology
- PSY 3081 - Cognitive Psychology
 - OR PSY 3082 - Mind, Brain, and Behavior
 - OR PSY 3056 - Social Psychology
 - OR PSY 3067 - The Psychology of Motivation and Learning
- This concentration includes computational work in one of Baruch's psychology laboratories. For that reason, it requires more courses than other concentrations. The psychology mentors are Dr. Tatiana Emmanouil and Dr. Jennifer Mangels.
- Independent study at one of the relevant psychology labs is also required for this concentration.
- This concentration requires one CS elective.

FINMATH-BS - Financial Mathematics

Overview

Institution Code

BAR01

Official Name of Program

Financial Mathematics

Plan Code

FINMATH-BS

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

Degree Designation

BS - Bachelor of Science

MTH-BA - Mathematics

Overview

Institution Code

BAR01

Official Name of Program

Mathematics

Plan Code

MTH-BA

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

Degree Designation

BA - Bachelor of Arts

Requirements

Simple Requisites

Degree Requirements - Bachelor of Science

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 120 credits

AND

Earn at least 30 credits

In Residence at Baruch College

AND

Earn at least 90 credits

Liberal Arts and Sciences Content Courses

AND

Earn a minimum GPA of 2.0

Degree Requirements - Bachelor of Arts

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 120 credits

AND

Earn at least 30 credits

In Residence at Baruch College

AND

Earn at least 90 credits

Liberal Arts and Sciences Content Courses

AND

Earn a minimum GPA of 2.0

General Education Requirements - Required Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

General Education Requirements - Required Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Required Core English Composition

Complete ALL of the following Courses:

- ENG 2100 - Writing I
OR ENG 2100T - Writing I
OR ENG 2100H - Honors Writing I
- ENG 2150 - Writing II
OR ENG 2150T - Writing II
OR ENG 2150H - Honors - Writing II

A course in this area must meet all of the following learning outcomes. A student will:

- Read and listen critically and analytically, including identifying an argument's major assumptions and assertions and evaluating its supporting evidence.
- Write clearly and coherently in varied, academic formats (such as formal essays, research papers, and reports) using standard English and appropriate technology to critique and improve one's own and others' texts.
- Demonstrate research skills using appropriate technology, including gathering, evaluating, and synthesizing primary and secondary sources.
- Support a thesis with well-reasoned arguments, and communicate persuasively across a variety of contexts, purposes, audiences, and media.
- Formulate original ideas and relate them to the ideas of others by employing the conventions of ethical attribution and citation.

AND

Required Core Mathematical and Quantitative Reasoning

Complete at least 1 of the following courses:

- MTH 2140 - Mathematics and Quantitative Reasoning
- MTH 2160 - Ideas in Mathematics and Their Applications
- MTH 2003 - Precalculus and Elements of Calculus 1A
- MTH 2009 - Precalculus
OR MTH 2009T - Precalculus
- MTH 2205 - Precalculus and Elements of Calculus 1B
- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 2610 - Calculus I
- MTH 2001 - Pre-Calculus
- PSY 2100 - Statistics for Social Science
- BIO 2100 - Biostatistics
OR ENV 2100 - Biostatistics
- STA 2000 - Business Statistics I
- MTH 2001 - Pre-Calculus
- MTH 2600 - Calculus I with Trigonometry Refresher

A course in this area must meet all of the following learning outcomes. A student will:

- Interpret and draw appropriate inferences from quantitative representations, such as formulas, graphs, or tables.
- Use algebraic, numerical, graphical, or statistical methods to draw accurate conclusions and solve mathematical problems.
- Represent quantitative problems expressed in natural language in a suitable mathematical format.
- Effectively communicate quantitative analysis or solutions to mathematical problems in written or oral form.
- Evaluate solutions to problems for reasonableness using a

Required Core English Composition

Complete ALL of the following Courses:

- ENG 2100 - Writing I
OR ENG 2100T - Writing I
OR ENG 2100H - Honors Writing I
- ENG 2150 - Writing II
OR ENG 2150T - Writing II
OR ENG 2150H - Honors - Writing II

A course in this area must meet all of the following learning outcomes. A student will:

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- MTH 2160 - Ideas in Mathematics and Their Applications
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- MTH 2009 - Precalculus
OR MTH 2009T - Precalculus
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- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 2610 - Calculus I
- MTH 2001 - Pre-Calculus
- PSY 2100 - Statistics for Social Science
- BIO 2100 - Biostatistics
OR ENV 2100 - Biostatistics
- STA 2000 - Business Statistics I
- MTH 2001 - Pre-Calculus
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- Use algebraic, numerical, graphical, or statistical methods to draw accurate conclusions and solve mathematical problems.
- Represent quantitative problems expressed in natural language in a suitable mathematical format.
- Effectively communicate quantitative analysis or solutions to mathematical problems in written or oral form.
- Evaluate solutions to problems for reasonableness using a

variety of means, including informed estimation.

- Apply mathematical methods to problems in other fields of study.

The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the Mathematical and Quantitative Reasoning requirement of the Pathways curriculum without appeal: MTH 2000; MTH 2300; MTH 2300; MTH 2301; MTH 2100; MTH 2150; MTH 2006; MTH 2206; and MTH 2630. MTH 3010 Calculus II is considered a STEM Variant course, and may also be used to satisfy this requirement without appeal.

**Zicklin students should not take MTH 2140 or MTH 2160, which do not meet the mathematics requirement for the BBA degree. Zicklin students must complete a course in precalculus (and must satisfy the prerequisites for whichever precalculus course they take). Therefore, these students usually take MTH 2003 or MTH 2009. Students who enter Baruch with a calculus placement (MTH 2207 or MTH 2610) may take STA 2000 to satisfy their Pathways requirement.

MTH 2140 and MTH 2160 are not appropriate for students within the Weissman School of Arts and Sciences whose major requires a statistics course or additional math courses (such majors include: Actuarial Science, Biological Sciences, Computer Science, Economics, Financial Mathematics, Natural Sciences ad hoc, and Statistics).

AND

Complete at least 1 courses in the following course sets:

- Required Core Life and Physical Sciences

A course in this area must meet all of the following learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a life or physical science.
- Apply the scientific method to explore natural phenomena, including hypothesis development, observation, experimentation, measurement, data analysis, and data presentation.
- Use the tools of a scientific discipline to carry out collaborative laboratory investigations.
- Gather, analyze, and interpret data and present it in an effective written laboratory or fieldwork report.
- Identify and apply research ethics and unbiased assessment in gathering and reporting scientific data.

NOTE: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the **Life and Physical Sciences** or the **Scientific World** requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the **Life and Physical Sciences** or the **Scientific World** requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO 2010 Principles of Biology I

variety of means, including informed estimation.

- Apply mathematical methods to problems in other fields of study.

The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the Mathematical and Quantitative Reasoning requirement of the Pathways curriculum without appeal: MTH 2000; MTH 2300; MTH 2300; MTH 2301; MTH 2100; MTH 2150; MTH 2006; MTH 2206; and MTH 2630. MTH 3010 Calculus II is considered a STEM Variant course, and may also be used to satisfy this requirement without appeal.

**Zicklin students should not take MTH 2140 or MTH 2160, which do not meet the mathematics requirement for the BBA degree. Zicklin students must complete a course in precalculus (and must satisfy the prerequisites for whichever precalculus course they take). Therefore, these students usually take MTH 2003 or MTH 2009. Students who enter Baruch with a calculus placement (MTH 2207 or MTH 2610) may take STA 2000 to satisfy their Pathways requirement.

MTH 2140 and MTH 2160 are not appropriate for students within the Weissman School of Arts and Sciences whose major requires a statistics course or additional math courses (such majors include: Actuarial Science, Biological Sciences, Computer Science, Economics, Financial Mathematics, Natural Sciences ad hoc, and Statistics).

AND

Complete at least 1 courses in the following course sets:

- Required Core Life and Physical Sciences

A course in this area must meet all of the following learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a life or physical science.
- Apply the scientific method to explore natural phenomena, including hypothesis development, observation, experimentation, measurement, data analysis, and data presentation.
- Use the tools of a scientific discipline to carry out collaborative laboratory investigations.
- Gather, analyze, and interpret data and present it in an effective written laboratory or fieldwork report.
- Identify and apply research ethics and unbiased assessment in gathering and reporting scientific data.

NOTE: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the **Life and Physical Sciences** or the **Scientific World** requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the **Life and Physical Sciences** or the **Scientific World** requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO 2010 Principles of Biology I

Baruch College

CHM 2003 General Chemistry I
PHY 2003 General Physics I
PHY 3010 Quantitative Physics I

Additional Comments:

The Department of Natural Sciences offers "stand-alone" sections of their Pathways courses (BIO 1011L, BIO 1012, etc.), specifically designed for students who are only required to complete one half of the paired courses – lecture or lab, not both. For example, a student who transfers a **Scientific World** course, may satisfy the **Life and Physical Sciences** requirement with a stand-alone section of a Natural Sciences course in that category. These stand-alone sections are not open to students who enter Baruch as freshmen or to transfer students who have satisfied neither the **Life & Physical Sciences** nor the **Scientific World** requirement at the time they entered Baruch. Please consult the Department of Natural Sciences to request registration permission for any of their stand-alone sections (17 Lexington Ave, room 506; 646-660-6200)

General Education Requirements - Flexible Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 courses in the following course sets:

- Flexible Core World Cultures and Global Issues

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring world cultures or global issues, including, but not limited to, anthropology, communications, cultural studies, economics, ethnic studies, foreign languages (building upon previous language acquisition), geography, history, political science, sociology, and world literature.
- Analyze culture, globalization, or global cultural diversity, and describe an event or process from more than one point of view.
- Analyze the historical development of one or more non-U.S. societies.
- Analyze the significance of one or more major movements that have shaped the world's societies.
- Analyze and discuss the role that race, ethnicity, class, gender, language, sexual orientation, belief, or other forms of social differentiation play in world cultures or societies.
- Speak, read, and write a language other than English, and use that language to respond to cultures other than one's own.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core U.S. Experience in Its Diversity

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the U.S. experience in its diversity, including, but not limited to

CHM 2003 General Chemistry I
PHY 2003 General Physics I
PHY 3010 Quantitative Physics I

Additional Comments:

The Department of Natural Sciences offers "stand-alone" sections of their Pathways courses (BIO 1011L, BIO 1012, etc.), specifically designed for students who are only required to complete one half of the paired courses – lecture or lab, not both. For example, a student who transfers a **Scientific World** course, may satisfy the **Life and Physical Sciences** requirement with a stand-alone section of a Natural Sciences course in that category. These stand-alone sections are not open to students who enter Baruch as freshmen or to transfer students who have satisfied neither the **Life & Physical Sciences** nor the **Scientific World** requirement at the time they entered Baruch. Please consult the Department of Natural Sciences to request registration permission for any of their stand-alone sections (17 Lexington Ave, room 506; 646-660-6200)

General Education Requirements - Flexible Core

Type

Completion Requirement

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- Analyze culture, globalization, or global cultural diversity, and describe an event or process from more than one point of view.
- Analyze the historical development of one or more non-U.S. societies.
- Analyze the significance of one or more major movements that have shaped the world's societies.
- Analyze and discuss the role that race, ethnicity, class, gender, language, sexual orientation, belief, or other forms of social differentiation play in world cultures or societies.
- Speak, read, and write a language other than English, and use that language to respond to cultures other than one's own.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core U.S. Experience in Its Diversity

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the U.S. experience in its diversity, including, but not limited to

experience in its diversity, meaning, but not limited to, anthropology, communications, cultural studies, economics, history, political science, psychology, public affairs, sociology, and U.S. literature.

- Analyze and explain one or more major themes of U.S. history from more than one informed perspective.
- Evaluate how indigenous populations, slavery, or immigration have shaped the development of the United States.
- Explain and evaluate the role of the United States in international relations.
- Identify and differentiate among the legislative, judicial, and executive branches of government and analyze their influence on the development of U.S. democracy.
- Analyze and discuss common institutions or patterns of life in contemporary U.S. society and how they influence, or are influenced by, race, ethnicity, class, gender, sexual orientation, belief, or other forms of social differentiation.

Note: *HIS 1004 History of American Civilization I* is no longer offered at Baruch, but it does appear on students' transfer evaluations. This course may be used to satisfy the U.S. Experience in its Diversity requirement of the Pathways curriculum without appeal.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Creative Expression

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring creative expression, including, but not limited to, arts, communications, creative writing, media arts, music, and theater.
- Analyze how arts from diverse cultures of the past serve as a foundation for those of the present, and describe the significance of works of art in the societies that created them.
- Articulate how meaning is created in the arts or communications and how experience is interpreted and conveyed.
- Demonstrate knowledge of the skills involved in the creative process.
- Use appropriate technologies to conduct research and to communicate.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Individual and Society

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the relationship between the individual and society, including, but not limited to, anthropology, communications, cultural studies, history, journalism, philosophy, political science, psychology, public affairs, religion, and sociology.
- Examine how an individual's place in society affects

experience in its diversity, meaning, but not limited to, anthropology, communications, cultural studies, economics, history, political science, psychology, public affairs, sociology, and U.S. literature.

- Analyze and explain one or more major themes of U.S. history from more than one informed perspective.
- Evaluate how indigenous populations, slavery, or immigration have shaped the development of the United States.
- Explain and evaluate the role of the United States in international relations.
- Identify and differentiate among the legislative, judicial, and executive branches of government and analyze their influence on the development of U.S. democracy.
- Analyze and discuss common institutions or patterns of life in contemporary U.S. society and how they influence, or are influenced by, race, ethnicity, class, gender, sexual orientation, belief, or other forms of social differentiation.

Note: *HIS 1004 History of American Civilization I* is no longer offered at Baruch, but it does appear on students' transfer evaluations. This course may be used to satisfy the U.S. Experience in its Diversity requirement of the Pathways curriculum without appeal.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Creative Expression

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring creative expression, including, but not limited to, arts, communications, creative writing, media arts, music, and theater.
- Analyze how arts from diverse cultures of the past serve as a foundation for those of the present, and describe the significance of works of art in the societies that created them.
- Articulate how meaning is created in the arts or communications and how experience is interpreted and conveyed.
- Demonstrate knowledge of the skills involved in the creative process.
- Use appropriate technologies to conduct research and to communicate.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Individual and Society

A course in this area must meet *at least three* of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the relationship between the individual and society, including, but not limited to, anthropology, communications, cultural studies, history, journalism, philosophy, political science, psychology, public affairs, religion, and sociology.
- Examine how an individual's place in society affects

- Examine how an individual's place in society affects experiences, values, or choices.
- Articulate and assess ethical views and their underlying premises.
- Articulate ethical uses of data and other information resources to respond to problems and questions.
- Identify and engage with local, national, or global trends or ideologies, and analyze their impact on individual or collective decision-making.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Scientific World

A course in this area must meet at least three of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the scientific world, including, but not limited to: computer science, history of science, life and physical sciences, linguistics, logic, mathematics, psychology, statistics, and technology-related studies.
- Demonstrate how tools of science, mathematics, technology, or formal analysis can be used to analyze problems and develop solutions.
- Articulate and evaluate the empirical evidence supporting a scientific or formal theory.
- Articulate and evaluate the impact of technologies and scientific discoveries on the contemporary world, such as issues of personal privacy, security, or ethical responsibilities.
- Understand the scientific principles underlying matters of policy or public concern in which science plays a role.

Note: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the *Scientific World* or the *Life and Physical Sciences* requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the *Life and Physical Sciences* or the *Scientific World* requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO2010 Principles of Biology I Principles of Biology I
CHM2003 General Chemistry I General Chemistry I
PHY2003 General Physics I General Physics I
PHY3010 Quantitative Physics I Quantitative Physics I

AND

Complete at least 1 courses in the following course sets:

- Flexible Core - Any Area

Additional Comments:

- Examine how an individual's place in society affects experiences, values, or choices.
- Articulate and assess ethical views and their underlying premises.
- Articulate ethical uses of data and other information resources to respond to problems and questions.
- Identify and engage with local, national, or global trends or ideologies, and analyze their impact on individual or collective decision-making.

AND

Complete at least 1 courses in the following course sets:

- Flexible Core Scientific World

A course in this area must meet at least three of the following *additional* learning outcomes. A student will:

- Identify and apply the fundamental concepts and methods of a discipline or interdisciplinary field exploring the scientific world, including, but not limited to: computer science, history of science, life and physical sciences, linguistics, logic, mathematics, psychology, statistics, and technology-related studies.
- Demonstrate how tools of science, mathematics, technology, or formal analysis can be used to analyze problems and develop solutions.
- Articulate and evaluate the empirical evidence supporting a scientific or formal theory.
- Articulate and evaluate the impact of technologies and scientific discoveries on the contemporary world, such as issues of personal privacy, security, or ethical responsibilities.
- Understand the scientific principles underlying matters of policy or public concern in which science plays a role.

Note: The following courses are no longer offered at Baruch, but do appear on students' transfer evaluations. Any of these courses may be used to satisfy the *Scientific World* or the *Life and Physical Sciences* requirement of the Pathways curriculum without appeal: BIO 1003; BIO 1005; CHM 1000; ENV 1020; ENV 1021; and PHY 1003. One course may not be used to satisfy both requirements.

STEM VARIANT COURSES

The following courses may be used to satisfy either the *Life and Physical Sciences* or the *Scientific World* requirement (one course may not satisfy both requirements). Students who wish to pursue a major or minor in the Natural Sciences should satisfy the Pathways science requirements with STEM Variant courses.

BIO2010 Principles of Biology I Principles of Biology I
CHM2003 General Chemistry I General Chemistry I
PHY2003 General Physics I General Physics I
PHY3010 Quantitative Physics I Quantitative Physics I

AND

Complete at least 1 courses in the following course sets:

- Flexible Core - Any Area

Additional Comments:

Baruch College

The Flexible Core is made up of six courses, which must be taken in five different categories (numbered 4-8): **4) World Cultures and Global Issues;** **5) U.S. Experience in its Diversity;** **6) Creative Expression;** **7) The Individual and Society;** and **8) Scientific World.**

In fulfilling the six-course requirement, students may not take more than two courses from any one department, discipline, or interdisciplinary field.

Please note that some departments offer courses in more than one category...

- ANT and SOC courses are offered by the Department of Sociology and Anthropology;
- ART, MSC, and THE courses are offered by the Department of Fine and Performing Arts;
- BLS and LTS (*formerly HSP*) courses are offered by the Department of Black and Latino Studies;
- BIO, ENV, CHM, and PHY courses are offered by the Department of Natural Science; and
- Within the Pathways Flexible Core, POL and PAF (*formerly PUB*) are considered to be a single field.

All Flexible Core courses must meet the following three learning outcomes. A student will:

- Gather, interpret, and assess information from a variety of sources and points of view.
- Evaluate evidence and arguments critically or analytically.
- Produce well-reasoned written or oral arguments using evidence to support conclusions.

The learning goals or outcomes of each area within the Flexible Core are included above, followed by lists of applicable courses.

Baruch no longer restricts students to one course from any given discipline, department, or interdisciplinary field

General Education Requirements - College Option

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete at least 1 of the following courses:

- CMP 2800 - Great Works of Literature I
OR ENG 2800 - Great Works of Literature I
- CMP 2850 - Great Works of Literature II
OR ENG 2850 - Great Works of Literature II

AND

Earn at least 3 credits

4000-level CIC "capstone" course for one of the liberal arts minors

This course must be completed at Baruch.

AND

Earn at least 6 credits

Two 3000-level liberal arts courses towards the same liberal arts minor.

- *The liberal arts minor cannot be within the same area of study as the major. This is also true for students who are required to complete part of the minor for the College Option.*

The Flexible Core is made up of six liberal arts courses, which must be taken in five different categories (numbered 4-8):²⁷²

4) World Cultures and Global Issues

5) U.S. Experience in its Diversity

6) Creative Expression

7) The Individual and Society

²⁸8) Scientific World.

In fulfilling the six-course requirement, students may not take more than two courses from any one department, discipline, or interdisciplinary field.²

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Baruch no longer restricts students to one course from any given discipline, department, or interdisciplinary field.

General Education Requirements - College Option

Type

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- CMP 2800 - Great Works of Literature I
OR ENG 2800 - Great Works of Literature I
- CMP 2850 - Great Works of Literature II
OR ENG 2850 - Great Works of Literature II

AND

Earn at least 3 credits

4000-level CIC "capstone" course for one of the liberal arts minors

This course must be completed at Baruch.

AND

Earn at least 6 credits

Two 3000-level liberal arts courses towards the same liberal arts

Baruch College

- *There is no GPA stipulation attached to the College Option courses, but in order to graduate with the liberal arts minor, a student must have a grade point average of at least 2.00 in the three courses that make up his or her liberal arts minor.*

Additional Comments:

The College Option consists of as many as four courses (6, 9, or 12 credits), depending on how much coursework the student has completed at another institution.

Major Requirements - Overall

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 43 credits

AND

Earn at least 27 credits

A minimum of 60% of the credits in the major field must be completed at Baruch College. For this major a minimum of 27 credits must be taken at Baruch.

AND

Earn a minimum GPA of 2.0

Major Requirements - Mathematics Program Prerequisites

Type

Prerequisite

Complete at least 1 of the following:

Option 1

Fulfill ALL of the following requirements:

- Calculus AP Exam (BC) with a score of 4 or 5 (*transfers to Baruch as MTH2610 Calculus I and MTH3010 Calculus II*)

AND

Complete ALL of the following Courses:

- MTH 3050 - Calculus III and Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

OR

Option 2

Fulfill ALL of the following requirements:

- Calculus AP exam (AB) with a score of 4 or 5 (*transfers to Baruch as MTH2610 Calculus I*)

THE COLLEGE OPTION CONSISTS OF AS MANY AS FOUR COURSES (6, 9, OR 12 CREDITS), DEPENDING ON HOW MUCH COURSEWORK THE STUDENT HAS COMPLETED AT ANOTHER INSTITUTION.

- *The liberal arts minor cannot be within the same area of study as the major. This is also true for students who are required to complete part of the minor for the College Option.*
- *There is no GPA stipulation attached to the College Option courses, but in order to graduate with the liberal arts minor, a student must have a grade point average of at least 2.00 in the three courses that make up his or her liberal arts minor.*

Additional Comments:

The College Option consists of as many as four courses (6, 9, or 12 credits), depending on how much coursework the student has completed at another institution.

Major Requirements - Overall

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 34 credits

AND

Earn a minimum GPA of 2.0

Major Requirements - Program Prerequisites

Type

Prerequisite

Complete at least 1 of the following:

Earn at least 8 credits from the following:

- MTH 3010 - Calculus II
- MTH 3020 - Calculus III

OR

Earn at least 8 credits from the following:

- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus

OR

Earn at least 12 credits from the following:

- MTH 2610 - Calculus I
- MTH 3010 - Calculus II
- MTH 3020 - Calculus III

OR

Earn at least 12 credits from the following:

- MTH 2610 - Calculus I
- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus

AND

Complete ALL of the following Courses:

- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

OR

Option 3

Complete ALL of the following Courses:

- MTH 2610 - Calculus I
- MTH 3010 - Calculus II
- MTH 3050 - Calculus III and Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

OR

Option 4

Fulfill ALL of the following requirements:

Complete at least 1 of the following courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
OR MTH 2206 - Applied Calculus
- MTH 2207 - Elements of Calculus I and Matrix Algebra

AND

Complete ALL of the following Courses:

- MTH 3006 - Elements of Calculus II
- MTH 3030 - Elements of Calculus III
- MTH 3035 - Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

OR

Option 5

Fulfill ALL of the following requirements:

Complete ALL of the following Courses:

- MTH 2205 - Precalculus and Elements of Calculus 1B
AND MTH 2206 - Applied Calculus
- MTH 2207 - Elements of Calculus I and Matrix Algebra

AND

Complete ALL of the following Courses:

- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3050 - Calculus III and Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

OR

OR

Earn at least 12 credits from the following:

- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 3006 - Elements of Calculus II
- MTH 3030 - Elements of Calculus III

OR

Earn at least 12 credits from the following:

- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3020 - Calculus III

OR

Earn at least 12 credits from the following:

- MTH 2207 - Elements of Calculus I and Matrix Algebra
- MTH 3006 - Elements of Calculus II
- MTH 3007 - Infinite Series
- MTH 3050 - Calculus III and Vector Calculus

OR

Earn at least 10 credits from the following:

- MTH 2630 - Analytic Geometry and Calculus I
- MTH 3030 - Elements of Calculus III

Major Requirements - Required Courses

Type

Completion Requirement

Complete ALL of the following Courses:

- MTH 3300 - Algorithms, Computers, and Programming I
- MTH 4010 - Mathematical Analysis I
- MTH 4100 - Linear Algebra and Matrix Methods

Major Requirements - Elective Courses

Type

Completion Requirement

Fulfill ALL of the following requirements:

Earn at least 15 credits from the following:

- MTH 4000 - Bridge to Higher Mathematics
- MTH 4009 - Proof Writing for Mathematical Analysis
- MTH 4020 - Advanced Calculus II
- MTH 4030 - Topology
- MTH 4110 - Ordinary Differential Equations
- MTH 4115 - Numerical Methods for Differential Equations in Finance
- MTH 4120 - Introduction to Probability

Option 6

Complete ALL of the following Courses:

- MTH 2630 - Analytic Geometry and Calculus I
- MTH 3030 - Elements of Calculus III
- MTH 3035 - Vector Calculus
- MTH 4000 - Bridge to Higher Mathematics

As a preliminary requirement, students must complete the calculus requirement, which may be achieved by any one of the following six methods:

Additional Comments:

1. At least an overall B+ average is required for the calculus courses preceding MTH3020 Calculus III, MTH3030 Elements of Calculus III or MTH3050 Calculus III and Vector Calculus in each of the above options.
2. MTH3050 Calculus III and Vector Calculus may be replaced with MTH3020 Calculus III and MTH3035 Vector Calculus in any of the above options.
3. At least a B or better is required in: MTH3050 Calculus III and Vector Calculus; or MTH3020 Calculus III and MTH3035 Vector Calculus; or MTH3030 Elements of Calculus III and MTH3035 Vector Calculus as appropriate.
4. To gain official admission to the program students must complete MTH4000 Bridge to Higher Mathematics with a minimum grade of B.

Major Requirements - Pre-Weissman Core

Type

Completion Requirement

Fulfill ALL of the following requirements:

Complete ALL of the following Courses:

- COM 1010 - Speech Communication
- OR COM 1010H - Honors Speech Communications

AND

Foreign Language

Earn at least 6 credits

Students may not use proficiency and/or exemption exams to satisfy this requirement.

1st Foreign Language course (based on placement) and 2nd Foreign Language course (the same language as the 1st Foreign Language course)

Students who transfer language credits for languages not taught at Baruch College may use these credits for the foreign language requirement at Baruch College.

Placement Information:

Students should review the guidelines below, and see a faculty advisor in the Department of Modern Languages and Comparative Literature to confirm proper placement if they intend to continue a language to which they have prior exposure. Students have the option of taking a placement exam to ensure that they are placed at the correct level.

Please consult the [Department of Modern Languages and Comparative Literature](#) for assistance

- MTH 4120H - Honors- Introduction to Probability
- MTH 4125 - Introduction to Stochastic Processes
- MTH 4130 - Mathematics of Data Analysis
- MTH 4135 - Computational Methods in Probability
- MTH 4140 - Graph Theory
- MTH 4145 - Mathematical Modeling
- MTH 4150 - Combinatorics
- MTH 4200 - Theory of Numbers
- MTH 4210 - Elements of Modern Algebra
- MTH 4220 - Introduction to Modern Geometry
- MTH 4230 - History of Mathematics
- MTH 4240 - Differential Geometry
- MTH 4300 - Algorithms, Computers and Programming II
- MTH 4310 - Methods of Numerical Analysis
- MTH 4315 - Introduction to Mathematical Logic
- MTH 4320 - Data Structures and Algorithms
- MTH 4330 - Introduction to Machine Learning
- MTH 4335 - Introduction to Natural Language Processing
- MTH 4421 - Actuarial Mathematics II
- MTH 4430 - Mathematics of Inferential Statistics
- MTH 4451 - Short-Term Insurance Mathematics
- MTH 4452 - Short-Term Insurance Mathematics II
- MTH 4500 - Introductory Financial Mathematics
- MTH 4600 - Data Analysis and Simulation for Financial Engineers
- MTH 5010 - Advance Calculus III
- MTH 5020 - Theory of Functions of a Complex Variable
- MTH 5030 - Theory of Functional of Real Variables
- MTH 5100 - Partial Differential Equations and Boundary Value Problems
- MTH 5500 - Stochastic Calculus for Finance

AND

Earn at least 3 credits from the following:

- Liberal Arts Electives

You may need to take additional Liberal Arts Courses to complete the required credits for your degree.

Major Requirements - Business Program Prerequisites

Type

Completion Requirement

Complete ALL of the following Courses:

- ACC 2101 - Principles of Accounting
- ECO 1001 - Micro-Economics
- ECO 1002 - Macro-Economics

Major Requirements - Required Courses

Type

Completion Requirement

Fulfill ALL of the following requirements:

Finance Courses

Complete ALL of the following Courses:

- FIN 3000 - Principles of Finance
- FIN 3610 - Corporate Finance

AND

Upper-level Mathematics Courses

Complete ALL of the following Courses:

- MTH 3300 - Algorithms, Computers, and Programming I
- MTH 4100 - Linear Algebra and Matrix Methods
OR MTH 3210 - A First Course in Linear Algebra
- MTH 4115 - Numerical Methods for Differential Equations in Finance
- MTH 4120 - Introduction to Probability
- MTH 4125 - Introduction to Stochastic Processes
- MTH 4130 - Mathematics of Data Analysis
- MTH 4300 - Algorithms, Computers and Programming II
- MTH 4500 - Introductory Financial Mathematics
- MTH 4600 - Data Analysis and Simulation for Financial Engineers
- MTH 5500 - Stochastic Calculus for Finance

Students who have completed MTH3120 Elementary Probability cannot enroll in MTH4120 Introduction to Probability. They must satisfy the probability requirement by registering for MTH4119 Multivariate Prob Dist as an independent study (please consult the Department of Mathematics).

Major Requirements - Electives

Type

Completion Requirement

Earn at least 3 credits from the following:

- Liberal Arts Electives

TRCSC-MIN - Computer Science

Overview

Institution Code

TRMTH-MIN - Mathematics

Overview

Institution Code

Baruch College

INSTITUTION CODE

BAR01

Official Name of Program

Computer Science

Plan Code

TRCSC-MIN

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

INSTITUTION CODE

BAR01

Official Name of Program

Mathematics

Plan Code

TRMTH-MIN

Career

Undergraduate

Department(s) Sponsoring Program

Mathematics

Requirements

Simple Requisites

Minor Requirements - Overall

Type

Completion Requirement

Earn at least 10 credits

Minor Requirements - Required Courses

Type

Completion Requirement

Required Courses

Complete ALL of the following Courses:

- MTH 3150 - Discrete Math: An Invitation to Computer Science
- MTH 3300 - Algorithms, Computers, and Programming I

Capstone Courses

Complete at least 1 of the following courses:

- MTH 4250 - Introduction to Cryptography
- MTH 4300 - Algorithms, Computers and Programming II
- MTH 4320 - Data Structures and Algorithms
- MTH 4325 - Programming Languages
- MTH 4330 - Introduction to Machine Learning
- MTH 4333 - Advanced Topics in Machine Learning
- MTH 4335 - Introduction to Natural Language Processing
- MTH 4350 - Computer Architecture
- MTH 4355 - Operating Systems
- MTH 4360 - Complexity and Computational Models
- MTH 4370 - Software Design
- MTH 4375 - Applied Data Privacy

Notes: No course may be used to satisfy requirements for both the computer science minor and a student's major.

* Students with credit for CIS 2300 cannot take MTH 3300. Students who cannot use MTH 3300 for the minor may either take two 4000-level courses from the list above, or one of the 4000-level courses together with MTH 4140 (Graph Theory).

** Students who have credit for CIS 3100 cannot take MTH 4300.

Requirements

Simple Requisites

Degree Requirements - Overall

Type

Completion Requirement

Earn at least 9 credits

Degree Requirements - Required Courses

Type

Completion Requirement

Fulfill ALL of the following requirements:

Required Courses

Complete at least 1 of the following courses:

- MTH 3006 - Elements of Calculus II
- MTH 3010 - Calculus II
- MTH 3020 - Calculus III
- MTH 3030 - Elements of Calculus III
- MTH 3050 - Calculus III and Vector Calculus

Degree Requirements - Electives and Captone

Type

Completion Requirement

Students must take two courses from the following list, with at least one of the courses being a 4000-level or higher capstone.

Complete at least 2 of the following courses:

- MTH 3020 - Calculus III
- MTH 3030 - Elements of Calculus III
- MTH 3050 - Calculus III and Vector Calculus
- MTH 3120 - Elementary Probability
- MTH 3150 - Discrete Math: An Invitation to Computer Science
- MTH 3210 - A First Course in Linear Algebra
- MTH 3300 - Algorithms, Computers, and Programming I
- MTH 4000 - Bridge to Higher Mathematics
- MTH 4010 - Mathematical Analysis I
- MTH 4020 - Advanced Calculus II
- MTH 4030 - Topology
- MTH 4100 - Linear Algebra and Matrix Methods
- MTH 4110 - Ordinary Differential Equations
- MTH 4115 - Numerical Methods for Differential Equations in Finance
- MTH 4120 - Introduction to Probability
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- MTH 4230 - History of Mathematics
- MTH 4240 - Differential Geometry
- MTH 4250 - Introduction to Cryptography
- MTH 4300 - Algorithms, Computers and Programming II
- MTH 4310 - Methods of Numerical Analysis
- MTH 4315 - Introduction to Mathematical Logic
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- MTH 4360 - Complexity and Computational Models
- MTH 4420 - Actuarial Mathematics I
- MTH 4421 - Actuarial Mathematics II
- MTH 4430 - Mathematics of Inferential Statistics
- MTH 4451 - Short-Term Insurance Mathematics
- MTH 4452 - Short-Term Insurance Mathematics II
- MTH 4500 - Introductory Financial Mathematics
- MTH 4600 - Data Analysis and Simulation for Financial Engineers
- MTH 5010 - Advance Calculus III
- MTH 5020 - Theory of Functions of a Complex Variable
- MTH 5030 - Theory of Functional of Real Variables
- MTH 5100 - Partial Differential Equations and Boundary Value Problems
- MTH 5500 - Stochastic Calculus for Finance
- MTH 4005, MTH 4009, MTH 4119, and MTH 4410 are not applicable to the mathematics minor
- None of these courses may be used to satisfy the capstone requirement: MTH 4230, MTH 4250, MTH 4300, MTH 4320, MTH 4325, MTH 4330, MTH 4350, MTH 4355, and MTH 4360

Courses

Baruch College

FSPM 1023 - Intermediate and College Algebra

Overview

Academic Institution

BAR01

Subject area

FSPM

Catalog Number

1023

Course Title

Intermediate and College Algebra

Department(s)

Mathematics

Description

This course is identical to MTH 1023 and is to be part of the college's summer immersion program for new students. It develops the algebraic topics that are needed for most basic quantitative courses at the college, including factoring, quadratics, linear equations, rates of change, rational and irrational expressions, functions and their graphs, non-linear systems of equations and related applications.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

0

Maximum Units:

0

MTH 123L - Mth 0123 Lect

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

123L

Course Title

Mth 0123 Lect

Department(s)

Mathematics

Description

Mth 0123 Lect

Academic Career

Undergraduate

Permission Type

No Special Consent Required

Credits

Minimum Units:

0

Maximum Units:

0

MTH 10 - Elementary Algebra I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

10

Course Title

Elementary Algebra I

Department(s)

Mathematics

Description

Elementary Algebra I

Academic Career

Undergraduate

Permission Type

No Special Consent Required

Credits

Minimum Units:

0

Maximum Units:

0

MTH 150B - Basic Math II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

150B

Course Title

Basic Math II

Department(s)

Mathematics

Description

Basic Math II

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

2

Maximum Units:

2

MTH 1023 - Intermediate and College Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

1023

Course Title

Intermediate and College Algebra

Department(s)

Mathematics

Description

This course develops the algebraic topics that are needed for most basic quantitative courses at the college, including factoring, quadratics, linear equations, rates of change, rational and irrational expressions, functions and their graphs, non-linear systems of equations and related applications. Note: Students who fail MTH 1023 in the spring semester have the option of taking courses in the Immersion Summer Program or in the regular summer session, and may complete coverage of the content of MTH 1023 by completing FSPM 0121 and either FSPM 1031 or MTH 1030. Consequently, either FSPM 1031 or MTH 1030 may be used to replace the F grade in MTH 1023.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

2

Maximum Units:

2

MTH 1030 - College Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

1030

Course Title

College Algebra

Department(s)

Mathematics

Description

This course develops the algebraic topics that are needed for most basic quantitative courses at the college, including linear equations, rates of change, rational expressions, functions and their graphs, Exponential and logarithmic functions, non-linear systems of equations and related applications. This course is not part of the Common Core Curriculum.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

2

Maximum Units:

2

MTH 1052 - Elementary Mathematics of Finance and Probability Theory

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

1052

Course Title

Elementary Mathematics of Finance and Probability Theory

Department(s)

Mathematics

Description

This course contains elementary mathematics concepts useful in finance, including sigma notation, geometric sums, compound interest simple annuities, general annuities, perpetuities, elementary probability theory, and an introduction to life annuities.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2000 - Pre-Calculus: College Algebra and Trigonometry

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2000

Course Title

Pre-Calculus: College Algebra and Trigonometry

Department(s)

Mathematics

Description

The course is designed to provide the student with the background needed for the student of calculus. Algebra topics to be discussed include algebraic functions and equations, exponential and logarithmic functions and equations, systems of linear equations, sequences, and series.

Trigonometric topics to be discussed include trigonometric functions, their graphs and applications, identities, equations, and inverse functions. Not open to students who have completed MTH 2001, 2004, 2005, 2006, 2010, 2011, 2030, or any mathematics course at the 3000 level or above.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 2001 - Pre-Calculus

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2001

Course Title

Pre-Calculus

Department(s)

Mathematics

Description

This course introduces the student to techniques of algebra, trigonometry, and other topics necessary for the study of calculus as well as quantitative courses in allied disciplines. Topics to be covered include advanced topics in algebra, analytic geometry of straight lines and conic sections, analytic trigonometry, exponential and logarithmic functions, and theory of algebraic equations. Not open to students who have completed MTH 2000, 2002T, 2003, 2004, 2005, 2009, 2009T, 2010, 2205, 2206, 2207, 2600, 2610, 2011, 2630, or any mathematics course at the 3000 level or above.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2002T - Pre-Calculus with Tutorial

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2002T

Course Title

Pre-Calculus with Tutorial

Department(s)

Mathematics

Description

This course contains most of the content of MTH 2001, with the exception of the theory of algebraic equations and operations with matrices. It also includes a tutorial covering topics from intermediate algebra. Not open to students who completed MTH 2000, 2001, 2004, 2005, 2006, 2007, 2010, 2011, 2030, or any math course at the 3000 level or above. This is an experimental course.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

2

Maximum Units:

2

MTH 2003 - Precalculus and Elements of Calculus 1A

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2003

Course Title

Precalculus and Elements of Calculus 1A

Department(s)

Mathematics

Description

This course will integrate material from precalculus with introductory topics from applied calculus, including a detailed study of functions, limits and continuity, the circle, tangent lines, rates of change, differentiation of algebraic functions, matrices and linear systems of equations. Applications from economics and finance will be included, and the use of the TI 89/92 calculator will be required. (Not open to students who have completed MTH 2000, 2001, 2205, 2206, 2207, 2610, 2630, or any course at the 3000 level or above.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2004 - College Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2004

Course Title

College Algebra

Department(s)

Mathematics

Description

College Algebra

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

Baruch College

MTH 2006W - Calculus Workshop

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2006W

Course Title

Calculus Workshop

Department(s)

Mathematics

Description

Calculus Workshop

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

0

Maximum Units:

0

MTH 2008 - Calculus Computer Laboratory

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2008

Course Title

Calculus Computer Laboratory

Department(s)

Mathematics

Description

This laboratory course is taken concurrently with sections of MTH 2610 (Calculus I) or MTH 2206 (Applied Calculus). The topics to be covered include an introduction to DOS, an introduction to the features of the software package, and the use of the program in such topics as functions, graphing, limits, continuity, differentiability, and introduction to integration.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 2009 - Precalculus

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2009

Course Title

Precalculus

Department(s)

Mathematics

Description

This course integrates material from pre-calculus with introductory topics from applied calculus, including a detailed study of functions, limits and continuity, the circle, tangent lines, rates of change, differentiation of algebraic functions, matrices, and systems of linear equations. Applications from economics and finance will be included and the use of the TI 89/92 calculator as well as Excel will be required. Students may receive credit for MTH 2009 or MTH 2003, not both. These courses may substitute for each other in the F-grade replacement policy. This course is also not open to students who completed MTH 2000, 2001, 2010, 2030, 2205, 2206, 2207, 2610, or 2630 or any mathematics course at the 3000-level or above. NOTE: The policy on repeating courses covers any combination of MTH 2009 and MTH 2003, e.g., one course taken three times, or a one-and-two combination. All combinations will be treated identically as three attempts.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2009T - Precalculus

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2009T

Course Title

Precalculus

Department(s)

Mathematics

Description

The course content is identical to MTH 2009: it integrates material from pre-calculus with introductory topics from applied calculus, including a detailed study of functions, limits, and continuity, the circle, tangent lines, rates of change, and differentiation of algebraic functions. Applications from economics and finance will be included and the use of the TI 89/92 calculator will be required. MTH 2009T is a six-hour course that includes additional instructional time with the mathematics professor. Students may receive credit for MTH 2009T, MTH 2009, or MTH 2003. These courses may substitute for each other in the F-grade replacement policy. This course is not open to students who completed MTH 2000, 2001, 2010, 2030, 2205, 2206, 2207, 2610, or 2630 or any mathematics course at the 3000-level or above. NOTE: The policy on repeating courses covers any combination of MTH 2009, MTH 2009T, and MTH 2003, e.g., one course taken three times, or a one-and-two combination. All combinations will be treated identically as three attempts.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2120 - Mathematics Appreciation

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2120

Course Title

Mathematics Appreciation

Department(s)

Mathematics

Description

This class is designed to introduce students with basic math skills to the beauty and importance of mathematics in the arts and sciences. Students will read and write about subjects that can be understood with a minimal amount of mathematics. Topics include the numbers pi and e, the divine proportion (the golden mean), Fibonacci sequence, patterns in art and nature, and different number systems. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2140 - Mathematics and Quantitative Reasoning

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2140

Course Title

Mathematics and Quantitative Reasoning

Department(s)

Mathematics

Description

This class is designed to introduce students to thinking critically and applying it to problem solving. It will review the necessary basic skills while examining problems in financial management, statistical reasoning, basic notions in probability, and applications in the arts, politics and business. (MTH 2140 is not recommended for students whose major requires a statistics course or another math course. It does not meet the BBA base curriculum math requirement.) This course is not open to students who completed any other mathematics course numbered 2000 or higher. MTH 2140 and MTH 2160 may substitute for each other in the F-replacement policy. The policy on repeating courses covers MTH 2140 and MTH 2160, e.g., one course taken three times, each course taken once, or a one-and-two combination. All combinations will be treated identically as three attempts.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2160 - Ideas in Mathematics and Their Applications

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2160

Course Title

Ideas in Mathematics and Their Applications

Department(s)

Mathematics

Description

This course is designed for the liberal arts student who wants to sample the intellectual breadth of mathematics. Topics are chosen which are representative of the following fields: number theory, infinity in mathematics, geometry and topology, modern physics, computer arithmetic, set theory, the history of mathematics, probability and statistics, and graph theory. Applications of the ideas are presented wherever feasible. Some possible topics include: primality, the nature and representation of numbers, the Euclidean Algorithm, numerical approximation, geometric and sequences and series, cardinality, the bridges of Königsberg problem, projective geometry, non-Euclidean geometry, relativity, binary arithmetic, symbolic logic, the life of a selected mathematician, games of chance, misusing data, planar graphs, and network analysis. (MTH 2160 is not recommended for students whose major requires a statistics course or another math course. It does not meet the BBA base curriculum math requirement.) This course is not open to students who completed any other mathematics course numbered 2000 or higher. MTH 2140 and MTH 2160 may substitute for each other in the F-replacement policy. The policy on repeating courses covers MTH 2140 and MTH 2160, e.g., one course taken three times, each course taken once, or a one-and-two combination. All combinations will be treated identically as three attempts.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2161T - Ideas In Math with Tutor Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2161T

Course Title

Ideas In Math with Tutor

Department(s)

Mathematics

Description

Ideas In Math with Tutor

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 2202 - Mathematics for Elementary and Early Childhood Education

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2202

Course Title

Mathematics for Elementary and Early Childhood Education

Department(s)

Mathematics

Description

This course develops elementary school mathematics from an advanced viewpoint. Emphasis is placed on developing reasoning and problem-solving skills. Topics included are operations and algorithms in the system of whole numbers, arithmetic operations, decimals, ratio, proportion, geometry, and measurement and applications for transfer to the elementary school curriculum.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2205 - Precalculus and Elements of Calculus 1B

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2205

Course Title

Precalculus and Elements of Calculus 1B

Department(s)

Mathematics

Description

This course will include the first and second derivative tests, optimization, exponential and logarithmic functions, Riemann sums, areas, antiderivatives and business applications. This course is not open to students who have completed MTH 2201, 2206, 2207 or 2610.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2207 - Elements of Calculus I and Matrix Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2207

Course Title

Elements of Calculus I and Matrix Algebra

Department(s)

Mathematics

Description

This course contains the calculus included in MTH 2003 and MTH 2205 and the matrix algebra found in MTH 2003. Topics to be discussed include algebra of matrices; inverses; linear systems of equations; Gaussian elimination; intuitive and geometric definitions of the limit; derivatives of algebraic, exponential and logarithmic functions; optimization problems; related rates; curve sketching and an introduction to integration. (Not open to students who have completed MTH 2003, 2009, 2205, or 2610.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 2301 - Concepts of Discrete Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2301

Course Title

Concepts of Discrete Mathematics

Department(s)

Mathematics

Description

This course covers various mathematics concepts relating to computer and information systems, including logic, truth tables, and logical arguments; Boolean algebras, Boolean functions, and elementary switching theory; mathematical induction, permutations, combinations, and binomial coefficients; matrix arithmetic and inversion and Gauss-Jordan reduction; and elementary concepts of graph theory. Not open to students who have completed MTH 2300 or any mathematics course numbered 4100 or above.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 2600 - Calculus I with Trigonometry Refresher

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2600

Course Title

Calculus I with Trigonometry Refresher

Department(s)

Mathematics

Description

Topics to be discussed include an introduction to functions including trigonometric functions, inequalities, limits, derivatives of functions, applications to geometry and elementary economic analysis, theory of maxima and minima, antiderivatives, integration of functions, applications to geometry, and fundamental theorem of calculus. A student may receive credit for only one of the following eight courses: MTH 2011, 2201, 2205, 2206, 2207, 2600, 2610, or 2630. MTH 2600 may substitute for all of these courses except for MTH 2630 in the F-grade replacement policy. The policy on repeating courses covers any combination of these courses, e.g., one course taken three times, or a one-and-two combination. All combinations will be treated identically as three attempts.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 2610 - Calculus I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

2610

Course Title

Calculus I

Department(s)

Mathematics

Description

Topics to be discussed include: functions, inequalities; limits; derivatives of functions; applications to geometry and elementary economic analysis; theory of maxima and minima; antiderivatives; integration of functions; applications to geometry; fundamental theorem of calculus. (A student may receive credit for only one of the following eight courses: MTH 2011, 2201, 2205, 2206, 2207, 2600, 2610, or 2630.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 2610H - Honors - Calculus I

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	2610H

Course Title

Honors - Calculus I

Department(s)

Mathematics

Description

Topics to be discussed include: functions, inequalities; limits; derivatives of functions; applications to geometry and elementary economic analysis; theory of maxima and minima; antiderivatives; integration of functions; applications to geometry; fundamental theorem of calculus. (A student may receive credit for only one of the following eight courses: MTH 2011, 2201, 2205, 2206, 2207, 2600, 2610, or 2630.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 2630H - Honors - Analysis Calculus I

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	2630H

Course Title

Honors - Analysis Calculus I

Department(s)

Mathematics

Description

Topics to be discussed include elements of plane analytic geometry; vectors in the plane and their properties; limits, derivatives, and integrals of functions of one variable; polynomial, rational, trigonometric, exponential, and logarithmic functions; theory of maxima and minima; and techniques of integration. (A student may receive credit for only one of the following six courses: MTH 2011, 2201, 2205, 2206, 2207, 2610, or 2630.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
5	5

MTH 2630 - Analytic Geometry and Calculus I

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	2630

Course Title

Analytic Geometry and Calculus I

Department(s)

Mathematics

Description

Topics to be discussed include elements of plane analytic geometry; vectors in the plane and their properties; limits, derivatives, and integrals of functions of one variable; polynomial, rational, trigonometric, exponential, and logarithmic functions; theory of maxima and minima; and techniques of integration. (A student may receive credit for only one of the following six courses: MTH 2011, 2201, 2205, 2206, 2207, 2610, or 2630.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
5	5

MTH 3006 - Elements of Calculus II

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	3006

Course Title

Elements of Calculus II

Department(s)

Mathematics

Description

Topics to be discussed include implicit differentiation, related rates, antiderivatives, definite integrals with applications, fundamental theorem of calculus, properties of trigonometric and inverse trigonometric functions, methods of integration, limits, and indeterminate forms. Not open to students who have completed MTH 2610, 2630, or 3010.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 3007 - Infinite Series

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3007

Course Title

Infinite Series

Department(s)

Mathematics

Description

This course concerns infinite series and various commonly used tests to determine their convergence or divergence, as well as power series expansions and Taylor and Maclaurin series representations of common functions. Students passing this course with a C- or better may proceed into MTH 3020; students passing with B+ or better may proceed into MTH 3050. Not open to students who have completed MTH 3010, or MTH 3030.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3010 - Calculus II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3010

Course Title

Calculus II

Department(s)

Mathematics

Description

Topics to be discussed include transcendental functions, techniques of integration, improper integration, numerical integration, Taylor polynomial, introduction to multivariable functions, Lagrange multipliers, and indefinite and definite multiple integrals. Not open to students who have completed MTH 2630, 3006, or 3030.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 3020 - Calculus III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3020

Course Title

Calculus III

Department(s)

Mathematics

Description

Topics to be discussed include conics, parametric equations, polar coordinates, vectors in the plane and three dimensional space, multivariable calculus; chain rules, implicit functions, change of variables in multiple integrals; limits, continuity, and differentiability. Not open to students who have completed MTH 3030, or MTH 3050.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 3030 - Elements of Calculus III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3030

Course Title

Elements of Calculus III

Department(s)

Mathematics

Description

Topics to be discussed include elements of solid geometry, level curves and surfaces, vectors in space and their properties, differential calculus of functions of several variables, infinite sequences and series, convergence tests, Taylor's formula with remainder, Taylor series expansion of functions, and analytic functions. Not open to students who have completed MTH 3010, MTH 3020, or MTH 3050. Not open to students who completed MTH 3007 with a grade of C- or better.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

5

Maximum Units:

5

MTH 3035 - Vector Calculus

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3035

Course Title

Vector Calculus

Department(s)

Mathematics

Description

This class concerns vector fields, line integrals and surface integrals. It focuses on four versions of the Fundamental Theorem of Calculus: the fundamental theorem of calculus for vector fields on curves; Green's theorem; Stokes's theorem; and the Divergence theorem.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3040 - Actuarial Seminar : R for Actuaries

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3040

Course Title

Actuarial Seminar : R for Actuaries

Department(s)

Mathematics

Description

This course is an introduction to R programming with applications in actuarial Science. Although many applications may be related to topics in probability, interest theory, insurance and risk modeling, and mathematical finance, no prior knowledge of these topics is presupposed.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

2

Maximum Units:

2

MTH 3050 - Calculus III and Vector Calculus

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3050

Course Title

Calculus III and Vector Calculus

Department(s)

Mathematics

Description

In this course, the primary goal is to study the geometry of change in two and three dimensional space. In particular, we use vectors to describe curves and surfaces in space mathematically, and to study the derivatives (rates of change) and integrals (average properties) of functions and vector fields that are defined on curves and surfaces. The unity between geometry and algebra is most succinctly expressed in the four versions of the Fundamental Theorem of Calculus that we study: the fundamental theorem of calculus for vector fields on curves, Green's theorem, Stokes' theorem, the Divergence theorem and applications. The emphasis will be on understanding the geometry behind numerous algebraic manipulations, while providing a bit more focus on mathematical concepts. Not open to students who have completed MTH 3020, MTH 3030, or MTH 3035.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 3100 - Selective Topics in Finite Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3100

Course Title

Selective Topics in Finite Mathematics

Department(s)

Mathematics

Description

This course introduces the fundamental and unifying concepts of discrete mathematics. Topics covered divide into three categories: (1) fundamental concepts of mathematics: definitions, proofs, sets, functions, elementary number theory; (2) discrete structures: graphs, counting; (3) discrete probability theory. The underlying goal is to teach students about careful mathematics: precisely stating assertions about well-defined mathematical objects and verifying those assertions using mathematically sound proofs. This course may not be used within the Mathematics major.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 3102 - Chaotic Dynamical Systems

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3102

Course Title

Chaotic Dynamical Systems

Department(s)

Mathematics

Description

Chaos arises in mathematics when arbitrarily small changes in initial data can lead to immense changes in results over a fixed time interval. This course studies these phenomena both in the abstract and by experimentally using the computer algebra systems running on the X terminals connected to an IBM RSC/6000 Powerstation. It begins with the iteration of polynomial maps of one real variable. Most important phenomena are already present in this simple situation. The iteration of quadratic functions of one complex variable will lead to the Mandelbrot set.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 3120 - Elementary Probability

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3120

Course Title

Elementary Probability

Department(s)

Mathematics

Description

This course will introduce the student to the basic elements of discrete probability including: sample spaces, rules of probability, independence, conditional probability, Bayes' Theorem, discrete multivariate distribution, covariance, correlation and various special discrete distributions. It will also cover a small amount of integral calculus and introduce the student to the Normal distribution and the Central Limit Theorem. This course is not open to students who have completed MTH 3020 or MTH 3030 or MTH 3050 or MTH 4120.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 3150 - Discrete Math: An Invitation to Computer Science

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3150

Course Title

Discrete Math: An Invitation to Computer Science

Department(s)

Mathematics

Description

Topics discussed in this course include propositional logic; sets and functions; elements of number theory, combinatorics, and probability; and graph theory. Instruction in proof writing will be provided. MTH 3150 is not open to students who have completed MTH 4000.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 3210 - A First Course in Linear Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3210

Course Title

A First Course in Linear Algebra

Department(s)

Mathematics

Description

This course provides an introduction to matrix methods and linear algebra for Euclidean spaces, with a particular focus on two-dimensional and three-dimensional Euclidean spaces. Topics include matrices, systems of linear equations, determinants, Gaussian elimination, vector spaces, linear independence and bases, linear transformations, orthogonality, eigenvalues, eigenvectors, and singular value decomposition. The course examines selected applications. Not open to students who are enrolled in or have completed MTH 4100.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 3300 - Algorithms, Computers, and Programming I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3300

Course Title

Algorithms, Computers, and Programming I

Department(s)

Mathematics

Description

The objective of this course is to provide the basic knowledge and experience necessary to use computers effectively by developing an understanding of the interplay between the computer, its associated languages, and the structured development of algorithms. Topics to be covered include computer organization and operation, data representation, algorithm development and specification, and programming languages and techniques. These topics are developed through the study and application of a higher-level language. MTH 3300 is not open to students who have completed CIS 3120. Students cannot get credit for both CIS 2300 and MTH 3300.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 3901 - Actuarial Science Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3901

Course Title

Actuarial Science Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in actuarial science. Students enhance their educational experience by working as actuarial science interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously and concurrently taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis. (Only one of the four courses may be taken each semester.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3902 - Actuarial Science Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3902

Course Title

Actuarial Science Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in actuarial science. Students enhance their educational experience by working as actuarial science interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously and concurrently taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis. (Only one of the four courses may be taken each semester.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3903 - Actuarial Science Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3903

Course Title

Actuarial Science Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in actuarial science. Students enhance their educational experience by working as actuarial science interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously and concurrently taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis. (Only one of the four courses may be taken each semester.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3904 - Actuarial Science Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3904

Course Title

Actuarial Science Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in actuarial science. Students enhance their educational experience by working as actuarial science interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously and concurrently taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis. (Only one of the four courses may be taken each semester.)

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3905 - Mathematics Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3905

Course Title

Mathematics Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in mathematics. Students enhance their educational experience by working as mathematics interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3906 - Mathematics Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3906

Course Title

Mathematics Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in mathematics. Students enhance their educational experience by working as mathematics interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3907 - Mathematics Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3907

Course Title

Mathematics Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in mathematics. Students enhance their educational experience by working as mathematics interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3908 - Mathematics Internship

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3908

Course Title

Mathematics Internship

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in mathematics. Students enhance their educational experience by working as mathematics interns. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3909 - Financial Mathematics Internship I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3909

Course Title

Financial Mathematics Internship I

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in financial mathematics. Students enhance their educational experiences by working as financial mathematics interns. Students keep journals describing each day's activities, duties, and responsibilities, with emphasis on how academic material is applied to carry them out. The journals will also emphasize new techniques, theories, and methods that students learn in the course of the internship. Weekly meetings with a faculty mentor are used to discuss the journals; and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with each student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. Another meeting is held near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3910 - Financial Mathematics Internship II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3910

Course Title

Financial Mathematics Internship II

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in financial mathematics. Students enhance their educational experiences by working as financial mathematics interns. Students keep journals describing each day's activities, duties, and responsibilities, with emphasis on how academic material is applied to carry them out. The journals will also emphasize new techniques, theories, and methods that students learn in the course of the internship. Weekly meetings with a faculty mentor are used to discuss the journals; and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with each student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. Another meeting is held near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3911 - Financial Mathematics Internship III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3911

Course Title

Financial Mathematics Internship III

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in financial mathematics. Students enhance their educational experiences by working as financial mathematics interns. Students keep journals describing each day's activities, duties, and responsibilities, with emphasis on how academic material is applied to carry them out. The journals will also emphasize new techniques, theories, and methods that students learn in the course of the internship. Weekly meetings with a faculty mentor are used to discuss the journals; and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with each student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. Another meeting is held near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3912 - Financial Mathematics Internship IV

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3912

Course Title

Financial Mathematics Internship IV

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in financial mathematics. Students enhance their educational experiences by working as financial mathematics interns. Students keep journals describing each day's activities, duties, and responsibilities, with emphasis on how academic material is applied to carry them out. The journals will also emphasize new techniques, theories, and methods that students learn in the course of the internship. Weekly meetings with a faculty mentor are used to discuss the journals; and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with each student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. Another meeting is held near the end of the semester to discuss the student's performance and to evaluate the experience. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3913 - Computer Science Internship I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3913

Course Title

Computer Science Internship I

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in computer science. Students enhance their educational experience by working as interns in software development, data science, or related areas. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. Only one of the department's four internship courses may be taken each semester. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3914 - Computer Science Internship II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3914

Course Title

Computer Science Internship II

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in computer science. Students enhance their educational experience by working as interns in software development, data science, or related areas. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. Only one of the department's four internship courses may be taken each semester. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3915 - Computer Science Internship III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3915

Course Title

Computer Science Internship III

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in computer science. Students enhance their educational experience by working as interns in software development, data science, or related areas. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. Only one of the department's four internship courses may be taken each semester. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 3916 - Computer Science Internship IV

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

3916

Course Title

Computer Science Internship IV

Department(s)

Mathematics

Description

This course consists of a coordinated and supervised internship in computer science. Students enhance their educational experience by working as interns in software development, data science, or related areas. Students keep a journal describing each day's activities, duties, and responsibilities with an emphasis on how material learned in academic courses previously taken are used to carry them out. The journal also will emphasize new techniques, theories, and methods that the student learns as part of the internship. Weekly meetings with a faculty mentor are used to discuss the journal and a final paper summarizing the experience is required. Students are supervised by a faculty mentor and professional staff in the organization. The faculty mentor meets with the student's immediate supervisor at least twice. The first meeting occurs near the beginning of the semester to discuss goals. There is another meeting near the end of the semester to discuss the student's performance and to evaluate the experience. Only one of the department's four internship courses may be taken each semester. This course is graded on a pass/fail basis.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 4000 - Bridge to Higher Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4000

Course Title

Bridge to Higher Mathematics

Department(s)

Mathematics

Description

This class introduces the fundamental and unifying concepts of the contemporary mathematics. Topics covered divide into four categories: 1) fundamental concepts of mathematics: definitions, proofs, sets, functions, elementary number theory; 2) discrete structures: graphs, counting; 3) discrete probability theory; 4) elements of analysis and topology. The underlying goal is to teach students about careful mathematics: precisely stating assertions about well-defined mathematical objects and verifying these assertions using mathematically sound proofs.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4005 - Problem-Solving Seminar

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4005

Course Title

Problem-Solving Seminar

Department(s)

Mathematics

Description

This course introduces the fundamental techniques for problem solving in mathematical and for creating/writing sound mathematical proofs. Topics covered include a variety of methods from disciplines such as number theory, classical inequalities, real analysis, combinatorial, Euclidean geometry and linear algebra that are needed for solving the challenging problems, brainteasers and puzzles that appear in worldwide mathematical contests and in job interviews for quantitative positions. The three-person team representing Baruch College at the annual William Lowell Putnam Mathematical Competition is chosen based on performance in this seminar. This course is graded on a pass/fail basis. It may not be used with in Mathematics major or minor.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4009 - Proof Writing for Mathematical Analysis

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4009

Course Title

Proof Writing for Mathematical Analysis

Department(s)

Mathematics

Description

This course is designed to prepare students who will be taking mathematics 4010 to understand, and compose for themselves, the kind of demanding proofs that occur in advanced calculus. This course may not be used in the mathematics minor. MTH 4009 is not open to students who have completed MTH 4000 or MTH 4010

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 4010 - Mathematical Analysis I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4010

Course Title

Mathematical Analysis I

Department(s)

Mathematics

Description

This course presents a rigorous treatment of the limit, continuity, differentiability, and integrability of a function of one variable. Other topics include real numbers and the axiom of continuity; convergence of a sequence of real numbers; elements of point set topology; and extensions and generalizations of the law of the mean.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4020 - Advanced Calculus II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4020

Course Title

Advanced Calculus II

Department(s)

Mathematics

Description

This course presents rigorous treatment of the limit, continuity, differentiability, and differential of a function of two or more variables. Other topics include integration; multiple, improper, line, and surface integrals; and implicit function theorems.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4030 - Topology

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4030

Course Title

Topology

Department(s)

Mathematics

Description

This class introduces the fundamental and unifying concepts of one of the pillars of contemporary mathematics, point-set, or general, topology. Topics covered are metric spaces, general topological spaces, connectedness, and compactness. The relationship between the topological concepts and the corresponding concepts in real numbers is emphasized.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4100 - Linear Algebra and Matrix Methods

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4100

Course Title

Linear Algebra and Matrix Methods

Department(s)

Mathematics

Description

Topics to be included are Gauss-Jordan reduction, linear independence, linear vector spaces, linear transformations, similarity of matrices, diagonalizable matrices, characteristic values and vectors, and symmetric matrices and quadratic forms.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4110 - Ordinary Differential Equations

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4110

Course Title

Ordinary Differential Equations

Department(s)

Mathematics

Description

Topics to be included are existence and uniqueness of solutions, first order equations, linear equations, series solutions of second order linear equations, Laplace transforms linear systems, boundary value problems, and numerical methods.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4115 - Numerical Methods for Differential Equations in Finance

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4115

Course Title

Numerical Methods for Differential Equations in Finance

Department(s)

Mathematics

Description

This course covers exact solutions of ordinary and partial differential equations, as well as numerical solutions to these differential equations using finite difference methods. The financial applications include the Black-Scholes model and corresponding formulas, as well as practical issues of computing implied volatilities for American and European options from market data. The course will provide students with practical numerical tools for financial derivatives valuation.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4119 - Multivariate Probability Distributions

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4119

Course Title

Multivariate Probability Distributions

Department(s)

Mathematics

Description

This class will examine the properties of bivariate and multivariate continuous probability distributions and their applications. Not open to students who have completed mathematics 4120.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 4120 - Introduction to Probability

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4120

Course Title

Introduction to Probability

Department(s)

Mathematics

Description

This course is an introduction to the theory of probability. Topics to be included are basic theorems of probability, permutations and combinations, binomial and multinomial theorems, random variables with densities, sequences of independent identically distributed random variables, method of moments, the moment-generating function, central limit theorem, and standard-type probability distributions. Not open to students who have completed MTH 3120.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4120H - Honors- Introduction to Probability

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4120H

Course Title

Honors- Introduction to Probability

Department(s)

Mathematics

Description

This course is an introduction to the theory of probability. Topics to be included are basic theorems of probability, permutations and combinations, binomial and multinomial theorems, random variables with densities, sequences of independent identically distributed random variables, method of moments, the moment-generating function, central limit theorem, and standard-type probability distributions. Not open to students who have completed MTH 3120.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4125 - Introduction to Stochastic Processes

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4125

Course Title

Introduction to Stochastic Processes

Department(s)

Mathematics

Description

This course is designed to introduce the undergraduate to the fundamental ideas of stochastic (or random) processes. Such processes are used in the study of mathematical models where there are elements of uncertainty and hence probabilistic quantities are introduced into the model. These models are found in fields such as the analysis of algorithms, the theory of queues, the pricing of stock options, financial mathematics, econometrics, linear programming, and biomathematics. The courses will cover the topics of Markov chains (discrete and continuous time), renewal theory, queuing theory, Brownian motion, and stationary processes. Applications of the various topics will also be discussed.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4130 - Mathematics of Data Analysis

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4130

Course Title

Mathematics of Data Analysis

Department(s)

Mathematics

Description

This course is an introduction to statistics with a focus on data analysis. Topics covered during the first half of the course include confidence intervals, hypothesis testing, and linear regression. The second half of the course concerns time-series with topics including exponential smoothing models, autoregressive and moving average models. Topics and methods in cluster analysis such as K-means cluster analysis and hierarchical cluster analysis will be covered near the end of the semester. Students are introduced to practical data analysis skills using statistical software such as SAS or MATLAB, or using the R programming language. Not open to students who have completed or are taking STA 3155 or STA 4155.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4135 - Computational Methods in Probability

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4135

Course Title

Computational Methods in Probability

Department(s)

Mathematics

Description

This course is an introduction to the numerical techniques of Monte Carlo simulation and recursion in applications where randomness occurs. Topics include: random number generators; generating discrete and continuous random variables; simulating systems with randomness; variance reduction techniques; optimization via recursion. Applications will be drawn from finance, insurance, and various other business settings.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4140 - Graph Theory

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4140

Course Title

Graph Theory

Department(s)

Mathematics

Description

This course covers basic topics in graph theory, including connectivity, Eulerian graphs, planarity, genus, Hamiltonicity, isomorphism, chromatic number, Ramsey numbers, and enumeration. These are followed by an introduction to networks with graph algorithms, including algorithms for a maximum matching in a graph and algorithms for maximum flow in a network.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4145 - Mathematical Modeling

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4145

Course Title

Mathematical Modeling

Department(s)

Mathematics

Description

Students will define and offer solutions to real-world problems and give both a written and oral presentation of their solutions. Examples of problems to be studied are scheduling, portfolio investment, and reliability of machinery. Some of the following mathematical techniques will be used in the formation of the solution to the problems: these include but are not limited to stochastic processes, linear algebra, differential equations, dynamical systems, and probability theory. The entire class will be expected to offer written critiques of the work presented by each of the students. Students may take this class either as part of a proposed major in operations research or as part of their major in mathematics.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4150 - Combinatorics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4150

Course Title

Combinatorics

Department(s)

Mathematics

Description

This course covers basic topics and techniques of combinatorics, including mathematical induction, the pigeonhole principle, permutations, combinations, distributions, binomial coefficients, generating functions, partitions, recurrence relations, inclusion-exclusion, block designs, and finite projective planes.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4200 - Theory of Numbers

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4200

Course Title

Theory of Numbers

Department(s)

Mathematics

Description

Topics to be included are divisibility, primes, unique factorization, arithmetic functions, perfect numbers, linear and quadratic congruencies, primitive roots, quadratic reciprocity, and Diophantine equations.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4210 - Elements of Modern Algebra

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4210

Course Title

Elements of Modern Algebra

Department(s)

Mathematics

Description

Topics to be included are rings, groups, integral domains, fields, vector spaces, axiomatic development of rational and complex numbers, equivalence relations, isomorphism's, mathematical induction, divisibility, congruence, and fundamental theorem of arithmetic.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4215 - Finite Fields, Algebraic Curves, and Applications

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4215

Course Title

Finite Fields, Algebraic Curves, and Applications

Department(s)

Mathematics

Description

This course introduces the student to the algebraic theory of finite fields and their applications in number theory, cryptography, and error correcting codes. These are the building blocks of all modern uses of algebra in the digital world for communication of information and for secure e-commerce.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4220 - Introduction to Modern Geometry

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4220

Course Title

Introduction to Modern Geometry

Department(s)

Mathematics

Description

Logical deficiencies in Euclidean geometry; rigorous axiomatic development of geometry; betweenness; distance; boundedness of distance; Euclidean and spherical geometry and their models; separation theorems; convexity; Pasch's postulate; geometric inequalities; the parallel postulate; hyperbolic, elliptic, and parabolic geometry; the Poincare model; selected topics.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4230 - History of Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4230

Course Title

History of Mathematics

Department(s)

Mathematics

Description

This course explores the evolution of mathematical ideas in the ancient period (approximately 2000 BCE to 1200 CE), and conclude with a study of the re-emergence of Mathematics in Europe between the fourteenth and sixteenth centuries, the origins of Calculus in the seventeenth century, and the nineteenth century development of rigor and abstraction. The emphasis will be on doing representative historically important mathematical problems, using both historical and modern techniques. In the ancient period mathematical contributions of Babylonian, Egyptian, Greek, Chinese, Indian, and Arab Mathematicians will be studied. Then re-emergence of Mathematics in Europe will be explored through the contributions of Fibonacci, Pacioli, Cardano, Vieta, Fermat, Descartes, Newton, and Leibniz.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4240 - Differential Geometry

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4240

Course Title

Differential Geometry

Department(s)

Mathematics

Description

This course is an introduction to the basic concepts in differential geometry. The topics to be discussed are the local and the global geometric properties of curves and of surfaces in three-dimensional Euclidean space: parametric representation of curves and surfaces, curvature, the normal, the tangent, and the binomial vectors of a curve, orientation, Jacobian matrix, the four-vertex problem, the index number, Gaussian curvature of a surface, and the Gauss map.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4250 - Introduction to Cryptography

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4250

Course Title

Introduction to Cryptography

Department(s)

Mathematics

Description

In this course, one becomes familiar with modern-day cryptography, including the algorithms and theory that lie behind them. Topics include stream and block ciphers, modular arithmetic, prime numbers and factorization, finite fields, private and public keys, discrete logarithm problem, collisions, attacks, cryptographic hash functions, Chinese remainder theorem, Pohlig-Hellman algorithm, factorization algorithms, primality testing, Diffie-Hellman key exchange, El Gamal public key cryptosystem, RSA public key cryptosystem, digital signatures, message integrity, and authenticated encryption.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4299 - Pointer-Based Data

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4299

Course Title

Pointer-Based Data

Department(s)

Mathematics

Description

Topics include: pointers and their applications to arrays, linked lists, stacks, binary search trees; recursive algorithms with pointer-based data structures; object-oriented programming with copy constructors, move constructors, copy assignment operators, and move assignment operators; binary representation of integers; floating point representation of real numbers. This course is not open to students who have completed CIS 4100 or MTH 4300. MTH 4299 is not applicable to the math minor or to the computer science minor.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 4300 - Algorithms, Computers and Programming II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4300

Course Title

Algorithms, Computers and Programming II

Department(s)

Mathematics

Description

This course is intended to be a sequel to MTH 3300 and focuses on the notion of classes, along with the concepts of pointers, inheritance, and polymorphism. The implementation of classes in a practical setting, such as scientific computing, quantitative finance, or Windows programming, will be part of the course. Not open to students who have completed CIS 4100. Students will not receive credit for both CIS 3100 and MTH 4300.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4310 - Methods of Numerical Analysis

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4310

Course Title

Methods of Numerical Analysis

Department(s)

Mathematics

Description

This course is concerned primarily with techniques for solving mathematical problems on digital computers. Topics to be included are: solutions of non-linear equations and systems; direct and indirect methods of solving linear systems; numerical differentiation and integration; polynomial interpolation. Not open to students who have completed MTH 4400.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4315 - Introduction to Mathematical Logic

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4315

Course Title

Introduction to Mathematical Logic

Department(s)

Mathematics

Description

This course will introduce the student to the basic notions of mathematical logic. Topics to be covered include the syntax and semantics of propositional and first order logic, the completeness and compactness theorems, computability theory (including the unsolvability of the halting problem and the concept of recursive function), and the Gödel incompleteness theorems. Time permitting; some elementary set theory will also be discussed.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4320 - Data Structures and Algorithms

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4320

Course Title

Data Structures and Algorithms

Department(s)

Mathematics

Description

In this course, one learns how to design efficient algorithms and becomes familiar with common real-world algorithms. Examples include algorithms for data compression, error correcting codes, common graph algorithms such as shortest paths and network flow, and more. The course also includes algorithm techniques such as divide-and-conquer, greedy algorithms, and dynamic programming.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 4322 - Computer Science Internship Seminar

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4322

Course Title

Computer Science Internship Seminar

Department(s)

Mathematics

Description

This course equips students with comprehensive knowledge and practical skills essential for technical internship interviews in roles like data scientist and software engineer. Covering advanced algorithms, model deployment, and real-world problem-solving, it ensures that students are proficient in the latest tools and techniques demanded by industry. Students attend lectures with the aim of reviewing the background material and learning fundamental strategies to successfully conduct a technical interview.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
3	3

MTH 4322 - Computer Science Internship Seminar

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4322

Course Title

Computer Science Internship Seminar

Department(s)

Mathematics

Description

This course equips students with comprehensive knowledge and practical skills essential for technical internship interviews in roles like data scientist and software engineer. Covering advanced algorithms, model deployment, and real-world problem-solving, it ensures that students are proficient in the latest tools and techniques demanded by industry. Students attend lectures with the aim of reviewing the background material and learning fundamental strategies to successfully conduct a technical interview.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
3	3

MTH 4325 - Programming Languages

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4325

Course Title

Programming Languages

Department(s)

Mathematics

Description

This course discusses features of programming languages, focusing on issues that are important for language design. Topics include syntax, parsing, and context-free grammars; scope and bindings; semantic analysis; evaluation of expressions and control flow; type systems; functions; object-oriented, functional, and scripting languages; and memory management.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 4330 - Introduction to Machine Learning

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4330

Course Title
Introduction to Machine Learning

Department(s)
Mathematics

Description
This course provides an introduction to Machine Learning, where students will learn both theory and application of this subject. Students will get exposure to a broad range of machine learning methods and hands-on practice with real data. Topics include linear and logistic regression, support vector machines, decision trees, dimensionality reduction, unsupervised learning, and neural networks.

Academic Career
Undergraduate

Liberal Arts
Yes

Permission Type
No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 4333 - Advanced Topics in Machine Learning

Overview

Academic Institution

BAR01

Subject area	Catalog Number
MTH	4333

Course Title
Advanced Topics in Machine Learning

Department(s)
Mathematics

Description
This course, together with MTH 4330 and MTH 4335, complete the Machine Learning corpus of knowledge in our Computer Science Major. Students will get exposure to a broad range of advanced machine learning methods and hands-on practice with real data. Topics include generative models for non-sequential data, graph neural networks, and reinforcement learning.

Academic Career
Undergraduate

Liberal Arts
Yes

Permission Type
No Special Consent Required

Credits

Minimum Units:	Maximum Units:
4	4

MTH 4335 - Introduction to Natural Language Processing

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4335

Course Title

Introduction to Natural Language Processing

Department(s)

Mathematics

Description

Recent developments in large language models, such as ChatGPT, have revolutionized how our society accommodates machine-based reasoning and analysis of text data, which is a prevalent form of unstructured data. This course will examine the ideas behind the implementation of these models. Topics include: basic techniques in text representation such as tokenization, tagging, chunking, co-occurrence and tf-idf; software for text analytics; sentiment analysis, categorization, and visualization of text data; basics of neural networks; word vectors and algorithms of word2vec and GloVe; the family of recurrent neural networks; attention and transformers; natural language generation.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4340 - Switching Theory

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4340

Course Title

Switching Theory

Department(s)

Mathematics

Description

This course covers logical design and switching circuit theory. Topics to be included are Boolean algebra, Boolean functions, Karnaugh maps, Quine-McCluskey tabulation method, gate implementation, sequential logic, synchronous and asynchronous sequential circuits, and applications.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4350 - Computer Architecture

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4350

Course Title

Computer Architecture

Department(s)

Mathematics

Description

In this course, one becomes familiar with main components of modern computer systems and learns to design efficient systems. Topics include Boolean circuits, the ALU, RISC, Instruction Set Architecture, arithmetic for computers, pipelining, the memory hierarchy, caching, address translation, memory protection, virtual memory, I/O, interrupts, branch prediction, dynamic scheduling, Tomasulo's algorithm, parallel computing, vector processors, and multicores.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4355 - Operating Systems

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4355

Course Title

Operating Systems

Department(s)

Mathematics

Description

In this course, one becomes familiar with the main components of operating systems, how to administer these components, and how to write code that interacts with them. Topics include processes; kernels; interrupts; system calls; interprocess communication; shared memory; message passing and ports; context switching; concurrency; threads; synchronization; mutual exclusion; mutex locks, semaphores, and monitors; CPU scheduling; multi-processor scheduling and load balancing; memory management; loading and linking; caching; paging; swapping; virtual memory; mass storage; I/O systems; file system; naming; directories; mounting; distributed file systems; networking; and security and protection.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4360 - Complexity and Computational Models

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4360

Course Title

Complexity and Computational Models

Department(s)

Mathematics

Description

Two fundamental questions arising in any problem are: Can this problem be solved using a given abstract machine? How much time and space are required to solve it? The theory of computational complexity provides tools for analyzing the minimal amount of computational resources that are needed for the algorithmic solution of a problem. In this course, we will discuss a variety of types of computational problems (decision, search, counting, and optimization) by introducing an array of complexity classes to capture problem types. We will use the notions of reduction and completeness to establish relationships between seemingly unrelated problems, classes, and resources.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4370 - Software Design

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4370

Course Title

Software Design

Department(s)

Mathematics

Description

Principles of design for creating large software projects are discussed. Topics include object-oriented paradigms; the software development life cycle; documenting requirements and specifications; testing; version control. Students will produce substantial software projects in teams and give written and oral reports detailing the software and its development.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4375 - Applied Data Privacy

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4375

Course Title

Applied Data Privacy

Department(s)

Mathematics

Description

Statistics and probability form the mathematical basis for handling data. This course introduces statistical techniques for the analysis of data sets with a focus on fundamental aspects of data privacy, which can be used to construct data-based solutions to problems that are encountered in the technology and government sectors. Topics discussed include experimental design; visualization techniques; and best practices for handling data, including encryption and statistical privacy. The ideas will be explored through statistical theory, case studies, and hands-on experimentation.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4375 - Applied Data Privacy

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4375

Course Title

Applied Data Privacy

Department(s)

Mathematics

Description

Statistics and probability form the mathematical basis for handling data. This course introduces statistical techniques for the analysis of data sets with a focus on fundamental aspects of data privacy, which can be used to construct data-based solutions to problems that are encountered in the technology and government sectors. Topics discussed include experimental design; visualization techniques; and best practices for handling data, including encryption and statistical privacy. The ideas will be explored through statistical theory, case studies, and hands-on experimentation.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4380 - Computer and Network Security

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4380

Course Title

Computer and Network Security

Department(s)

Mathematics

Description

This course offers a comprehensive introduction to the fundamentals of computer and network security, covering key principles and practices essential for safeguarding modern computing systems. Topics include the fundamentals of cybersecurity, cryptography, secure software development, system security, and network security. Students will learn to identify and mitigate threats, implement cryptographic techniques, and design secure architectures for web, mobile, and network platforms. The course emphasizes practical applications through hands-on labs using tools such as firewalls, intrusion detection systems, and VPNs. Emerging topics, including blockchain security, AI-based defenses, post-quantum cryptography, and IoT security, are introduced to address evolving cyber threats. Ethical considerations, legal frameworks, and global cybersecurity policies are integrated throughout the course. Students will gain foundational skills to apply these concepts in real-world scenarios, preparing them for careers in the dynamic field of cybersecurity. Students may not receive credit for both MTH 4380 and CIS 3550.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4380 - Computer and Network Security

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4380

Course Title

Computer and Network Security

Department(s)

Mathematics

Description

This course offers a comprehensive introduction to the fundamentals of computer and network security, covering key principles and practices essential for safeguarding modern computing systems. Topics include the fundamentals of cybersecurity, cryptography, secure software development, system security, and network security. Students will learn to identify and mitigate threats, implement cryptographic techniques, and design secure architectures for web, mobile, and network platforms. The course emphasizes practical applications through hands-on labs using tools such as firewalls, intrusion detection systems, and VPNs. Emerging topics, including blockchain security, AI-based defenses, post-quantum cryptography, and IoT security, are introduced to address evolving cyber threats. Ethical considerations, legal frameworks, and global cybersecurity policies are integrated throughout the course. Students will gain foundational skills to apply these concepts in real-world scenarios, preparing them for careers in the dynamic field of cybersecurity. Students may not receive credit for both MTH 4380 and CIS 3550.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4395 - Special Topics in Computer Science

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4395

Course Title

Special Topics in Computer Science

Department(s)

Mathematics

Description

Advanced study of selected topics in computer science that are not treated in regular courses.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 4400 - Finite Differences

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4400

Course Title

Finite Differences

Department(s)

Mathematics

Description

This course develops the theory of finite differences methods. Topics to be included are interpolations with equal and unequal intervals, Sheppard's Rules, interpolation of functions of several variables, central difference formulas, summation, approximate differentiation and integration with equal and unequal intervals, difference equations, iteration, and systems of equations. Not open to students who have completed MTH 4310.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4410 - Theory of Interest

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4410

Course Title

Theory of Interest

Department(s)

Mathematics

Description

This course provides a mathematical understanding of fundamental concepts offinancial mathematics and their applications in calculating values for various streams of cash flows. Topics to be discussed include measurements of various interest rates, equations of value, yield rates, noncontingent annuities, amortization schedules, sinking funds, bonds, interest rate swaps, and immunization.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4420 - Actuarial Mathematics I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4420

Course Title

Actuarial Mathematics I

Department(s)

Mathematics

Description

This course applies probability and the theory of interest to the prices of insurance products. Topics to be covered include survival distributions and life tables, life insurance (whole, term, endowment, deferred, and variable benefit insurance), life annuities, premiums (net and gross), and premium reserves (net and gross).

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4421 - Actuarial Mathematics II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4421

Course Title

Actuarial Mathematics II

Department(s)

Mathematics

Description

This course is a continuation of MTH 4420. Topics to be covered include multiple state and multiple decrement models, joint life and last survivor models, valuation theory for pension plans, non-diversifiable risk, profit testing for insurance portfolios, dividends, and participating and universal life insurance policies.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4430 - Mathematics of Inferential Statistics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4430

Course Title

Mathematics of Inferential Statistics

Department(s)

Mathematics

Description

This class provides a calculus-based introduction to inferential statistics. Basic concepts of random sampling and statistics including confidence intervals, bias, the method of moments, percentile matching, and sampling distribution estimators will be covered. Additional topics include hypothesis testing using concepts such as the t-test, Chi-square test, goodness-of-fit test, and the Neyman-Pearson Lemma; maximum likelihood estimators; Bayes estimators; sufficient and robust statistics; information criteria; and an introduction to inference on linear models.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4451 - Short-Term Insurance Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4451

Course Title

Short-Term Insurance Mathematics

Department(s)

Mathematics

Description

This course builds upon probability theory to introduce a variety of frequency and severity models, the steps involved in the modeling process, and procedures for carrying out these steps in solving business problems. The major topics to be studied are severity, frequency, and aggregate models and their modifications, as well as risk measures and the construction of empirical models for short-term insurance. This course is intended for actuarial science students preparing for the professional examination on the Construction and Evaluation of Actuarial Models (Exam C) given by the Casualty Actuarial Society and the Society of Actuaries.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4452 - Short-Term Insurance Mathematics II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4452

Course Title

Short-Term Insurance Mathematics II

Department(s)

Mathematics

Description

This course is a continuation of MTH 4451. The major topics to be covered include credibility procedures for estimating losses, and construction, estimation, and selection of parametric models. This course is intended for actuarial science students and includes topics appearing on the Short Term Actuarial Mathematics (Exam STAM) given by the Casualty Actuarial Society and the Society of Actuaries.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 4500 - Introductory Financial Mathematics

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4500

Course Title

Introductory Financial Mathematics

Department(s)

Mathematics

Description

This course is an introduction to mathematical methods used in finance and their applications. No prior knowledge of finance is assumed. Basic financial instruments such as forward and futures contracts, options, and bonds are introduced. The course is built around three major themes: (i) risk-free assets and the term structure of interest rates; (ii) Markowitz portfolio optimization and the Capital Asset Pricing Model; (iii) No Arbitrage principle and its applications including pricing and hedging of derivative securities in the context of the multi-period binomial model and its continuous analog, the BlackScholes model. Students are expected to use their knowledge of probability, single and multivariable calculus, and basic linear algebra to master mathematical finance theories and apply them in real world situations.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 5000 - Independent Study Math I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5000

Course Title

Independent Study Math I

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

3

MTH 4600 - Data Analysis and Simulation for Financial Engineers

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

4600

Course Title

Data Analysis and Simulation for Financial Engineers

Department(s)

Mathematics

Description

This "capstone" course for financial engineering majors brings to life their sophisticated mathematical background (in probability, stochastic processes, statistics, and financial mathematics) by connecting it to hands-on analysis of data and simulation techniques needed for real world financial engineering applications. Course topics include time series analysis, Markov Chain Monte Carlo methods, hidden Markov models, portfolio risk management. MTH 4600 can be used as an elective within the math minor, as a required course within the proposed financial engineering major, or as a general elective for the BA, BBA, or BS degrees. Prerequisite: MTH 4125, MTH 4130, MTH 4500.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 5000H - Hon - Independent Study Math I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5000H

Course Title

Hon - Independent Study Math I

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

3

Baruch College

MTH 5001 - Independent Study Math II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5001

Course Title

Independent Study Math II

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

4

MTH 5002 - Independent Study Math III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5002

Course Title

Independent Study Math III

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5001H - Hon Independent Study MTH II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5001H

Course Title

Hon Independent Study MTH II

Department(s)

Mathematics

Description

See the Department for description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

4

MTH 5003 - Independent Study Math IV

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5003

Course Title

Independent Study Math IV

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5004 - Independent Study Math V

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5004

Course Title

Independent Study Math V

Department(s)

Mathematics

Description

See Department for Description.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

1

Maximum Units:

1

MTH 5010 - Advance Calculus III

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5010

Course Title

Advance Calculus III

Department(s)

Mathematics

Description

Integration in n-space, implicit and inverse function theorems change of variables in multiple integrals, vector fields, line and surface integrals, theorems of Green, Stokes, and Gauss.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5020 - Theory of Functions of a Complex Variable

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5020

Course Title

Theory of Functions of a Complex Variable

Department(s)

Mathematics

Description

This course discusses the basic concepts in the theory of complex variables. The topics to be discussed will include the planar representation of the complex numbers, De Moivre's theorem, limits, Cauchy-Riemann equations, analytic functions, Taylor and Laurent series, integration, theory of residues, and introduction to Riemann surfaces.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5030 - Theory of Functional of Real Variables

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5030

Course Title

Theory of Functional of Real Variables

Department(s)

Mathematics

Description

Lebesgue measure and integration on the real line, differentiation of real functions and the relation with integration, classical L_p spaces.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5100 - Partial Differential Equations and Boundary Value Problems

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5100

Course Title

Partial Differential Equations and Boundary Value Problems

Department(s)

Mathematics

Description

This course is an introduction to partial differential equations. Topics to be included are first-order equations, classification of second-order equations, the method of separation of variables, and an introduction to Fourier series and transform methods.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 5220 - Probability Theory (Introduction to Probability)

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5220

Course Title

Probability Theory (Introduction to Probability)

Department(s)

Mathematics

Description

Basic theorems of probability, permutations and combinations, binomial and multinomial theorems, mathematical expectation, Stirling's theorem, central limit theorem, and standard type probability distributions.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 5500 - Stochastic Calculus for Finance

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

5500

Course Title

Stochastic Calculus for Finance

Department(s)

Mathematics

Description

This course is an introduction to stochastic calculus and its applications to modern finance. The core topics developed in the course are the Ito stochastic integral, Ito's formula, and basic stochastic differential equations. The course will provide students with indispensable tools for valuation of financial derivatives and for keeping up with developments in financial modeling. MTH 5500 can be used as an elective within the actuarial science and mathematics majors, as an elective within the mathematics minor, as a required course for the proposed financial engineering major, or as a general elective for the BA, BBA, or BS degrees. Prerequisite: MTH 4120, MTH 4125, MTH 4500.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

No Special Consent Required

Credits

Minimum Units:

4

Maximum Units:

4

MTH 6002H - Honors - MATH II

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

6002H

Course Title

Honors - MATH II

Department(s)

Mathematics

Description

The honors program in English is designed for the outstanding student. Each student conducts research in an area specific to the student's interest and works closely with a faculty advisor. Registration is by permission only from the department chairperson, honors committee designate, and the faculty sponsor.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

MTH 6001H - Honors - MATH I

Overview

Academic Institution

BAR01

Subject area

MTH

Catalog Number

6001H

Course Title

Honors - MATH I

Department(s)

Mathematics

Description

The honors program in English is designed for the outstanding student. Each student conducts research in an area specific to the student's interest and works closely with a faculty advisor. Registration is by permission only from the department chairperson, honors committee designate, and the faculty sponsor.

Academic Career

Undergraduate

Liberal Arts

Yes

Permission Type

Department Consent Required

Credits

Minimum Units:

3

Maximum Units:

3

